MASTER IN ASTROPHYSICS AND SPACE SCIENCE

Nicolaus Copernicus Astronomical Center of the Polish Academy of Sciences

Numerical Simulations of AGN Accretion Disks in PLUTO

CAMK PAN Summer Student Program

Fritz Ali Agildere

October 13, 2026

Overview

Numerical Setup
- Black Hole Adaptation
- Radiation Physics

Disk Classifier
- Passive Tracer
- Construction & Validation

Results
- Relaxation
- Steady State
- Density Waves

Discussion
- Limitations
- Outlook

Black Hole Adaptation

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  - Rate Capped above 1×10^9 K, Decoupled above $0.9\,c$ and Gas Floor at 1×10^4 K

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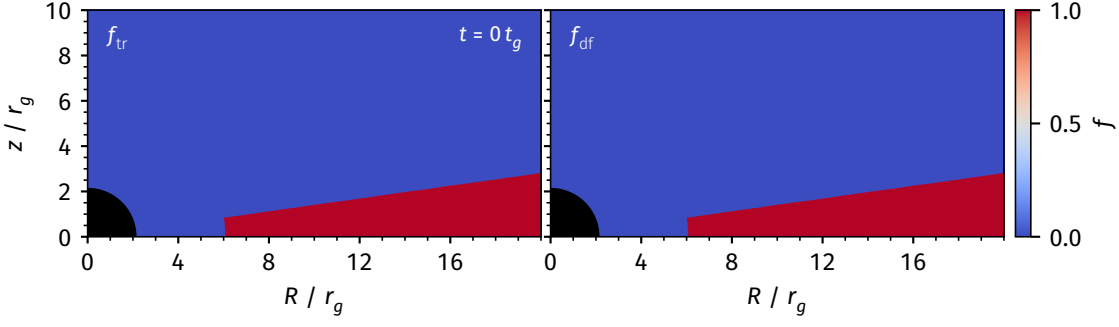
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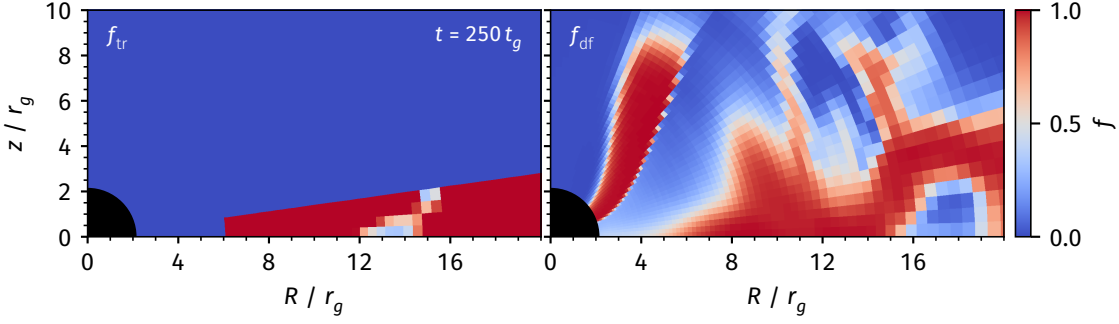
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  - Robust to Restarts by Writing Classifier State with Every Output

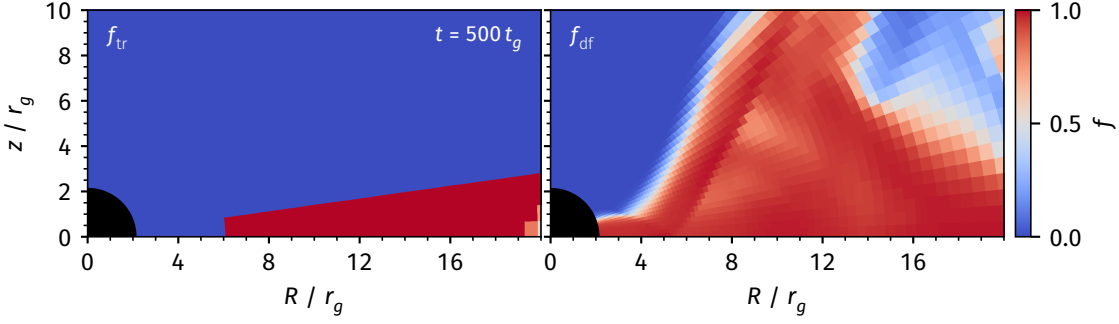
Radiation Enabled – Tracer & Classifier



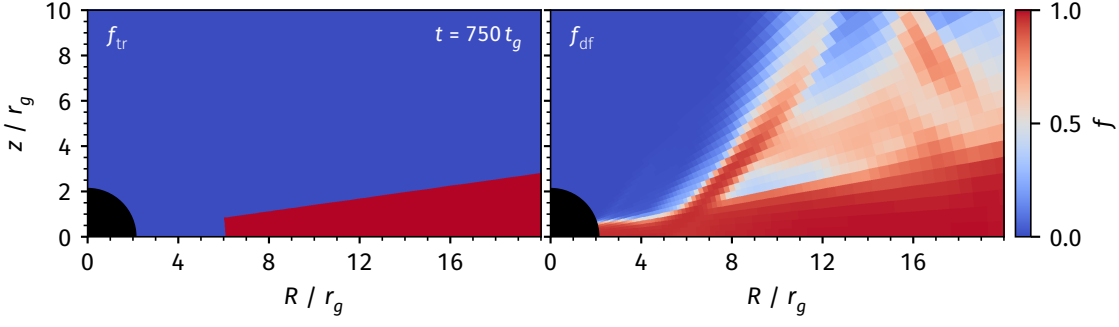
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  - Updated References as f and $1 - f$ Weighted Means via Moving Average with Weight $w = 1 - \exp(-\Delta t/\tau)$ and Time Constant $\tau = 6\, t_g$

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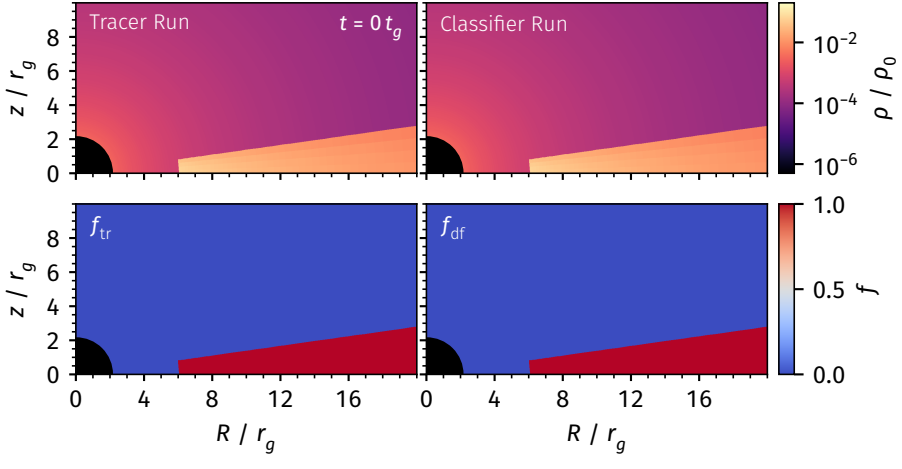
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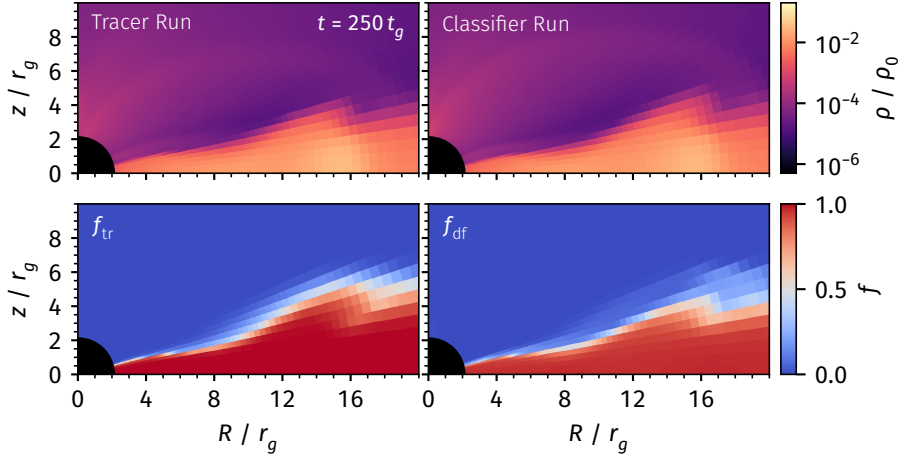
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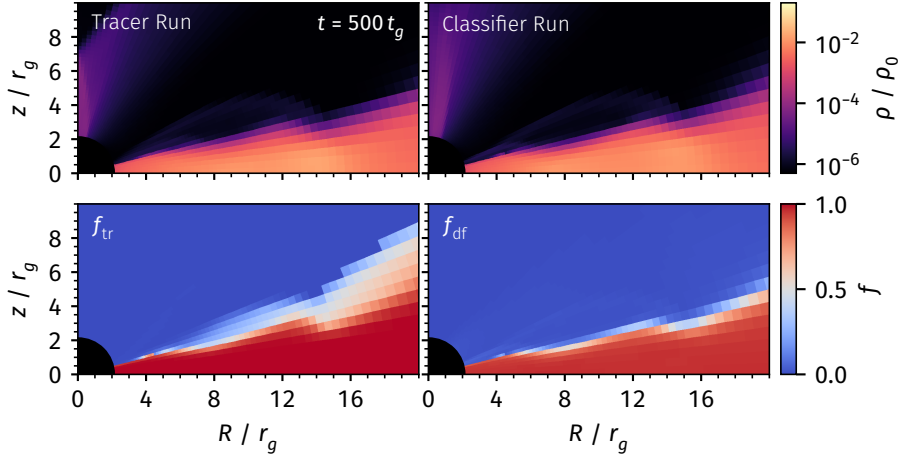
Validation – Hydrodynamics



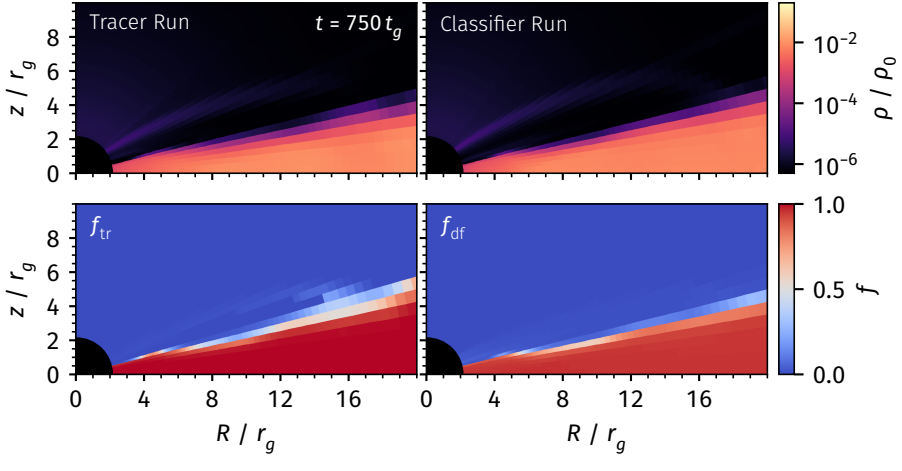
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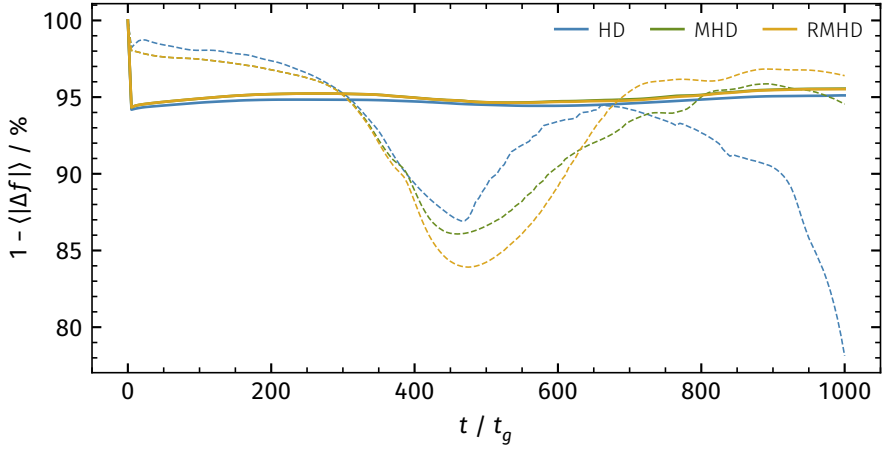
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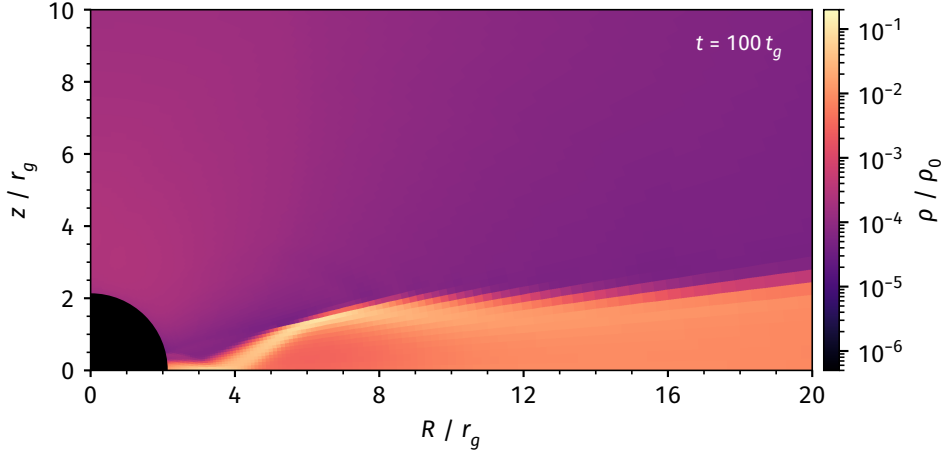
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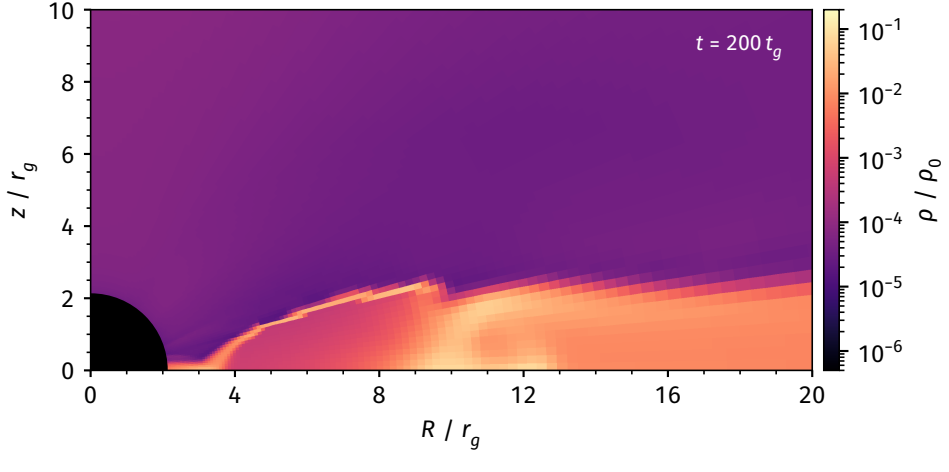
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  - Reached at $t = 23725$ for HR and $t = 9225$ for LR

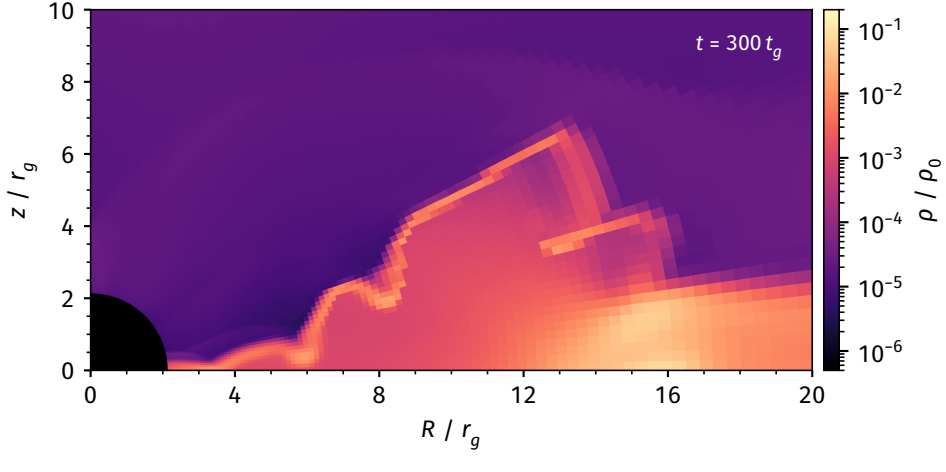
Early Evolution – Density



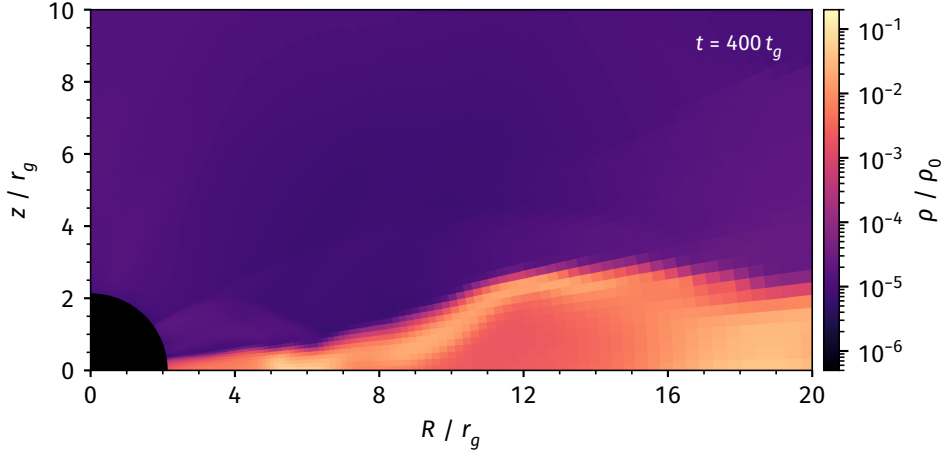
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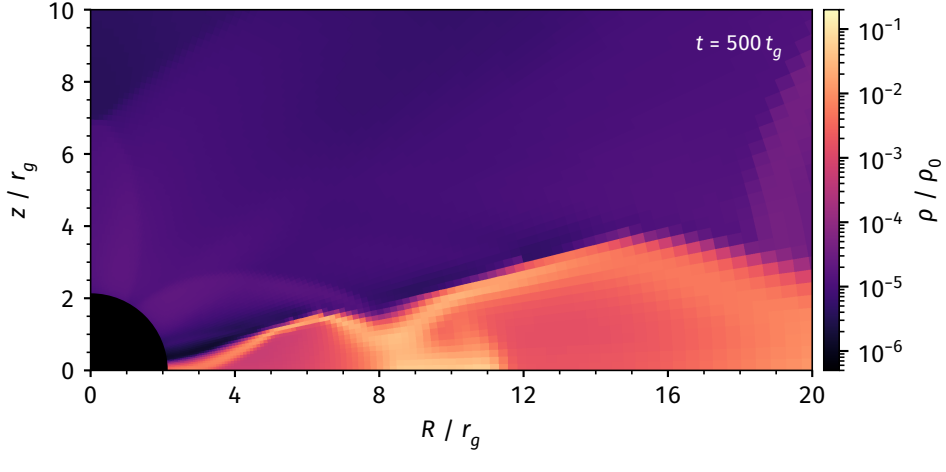
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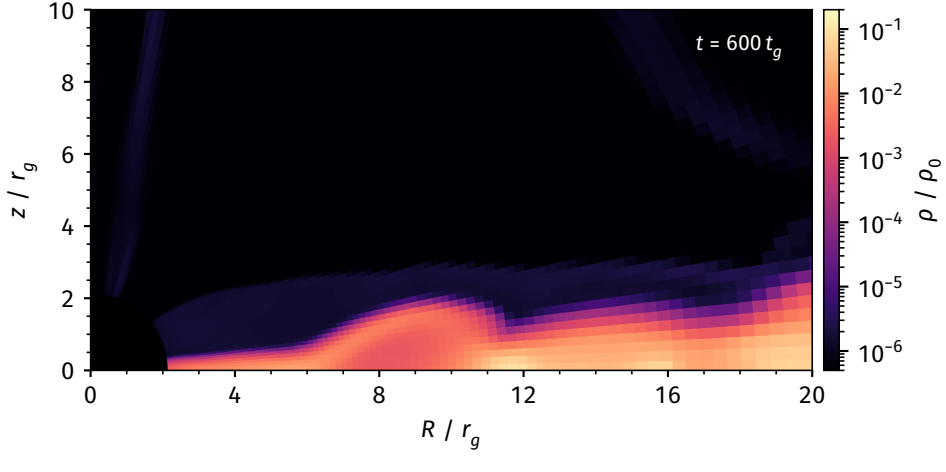
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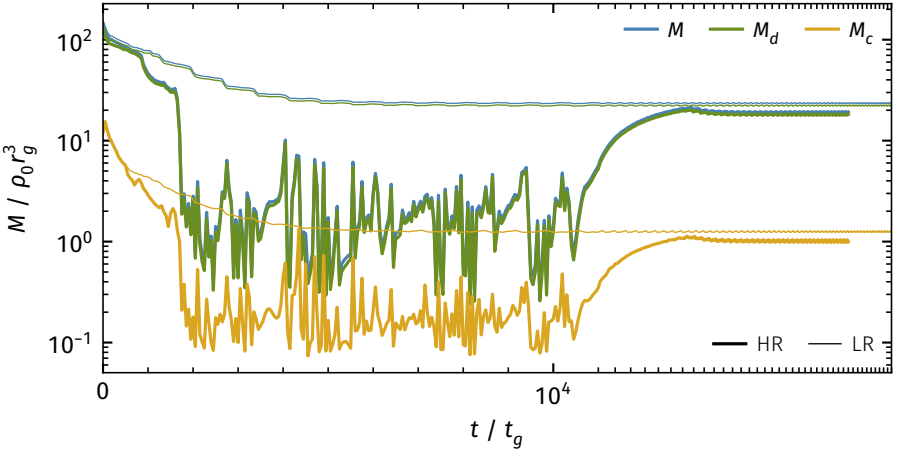
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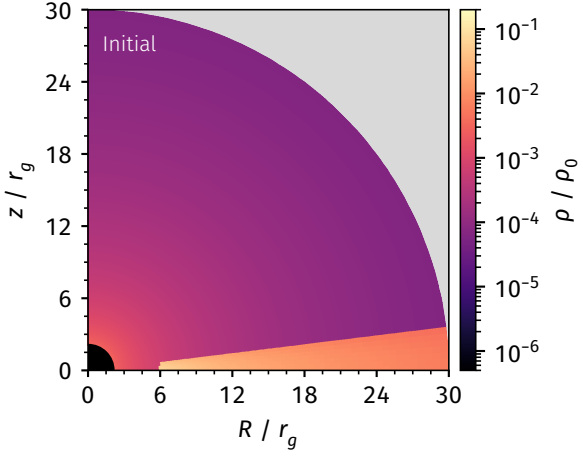
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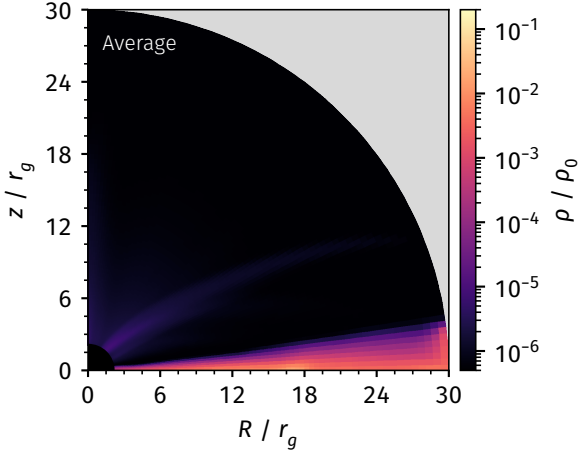
Relaxation – Mass



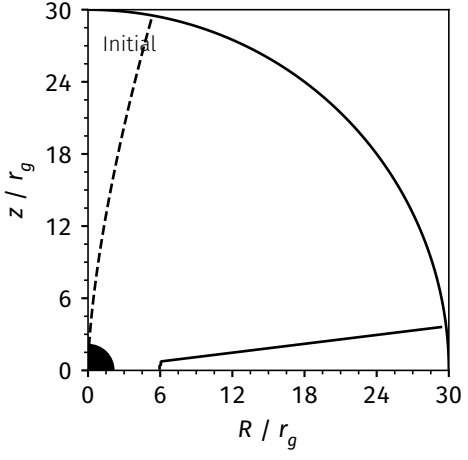
Steady State – Density



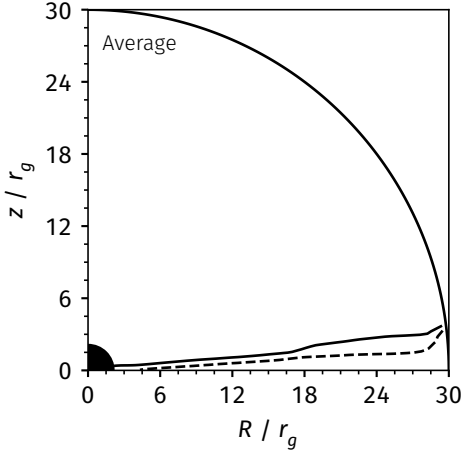
Steady State – Density



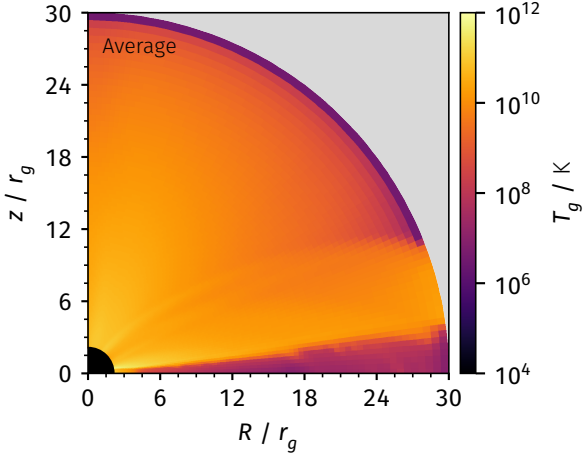
Steady State – Disk Surface



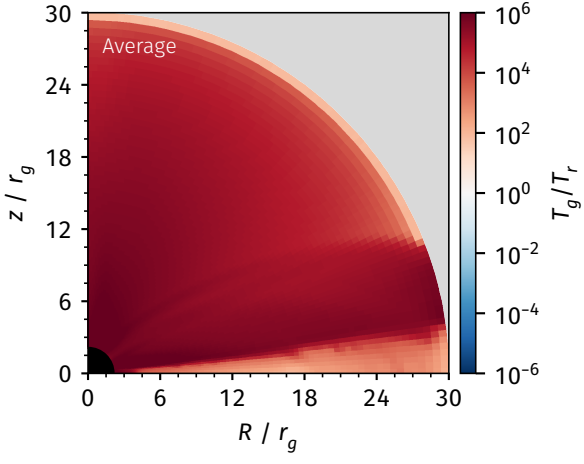
Steady State – Disk Surface



Steady State – Gas Temperature



Steady State – Temperature Ratio



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- Viscous Heating Balanced by Compton Cooling, Ratio of **0.8** to **0.9**

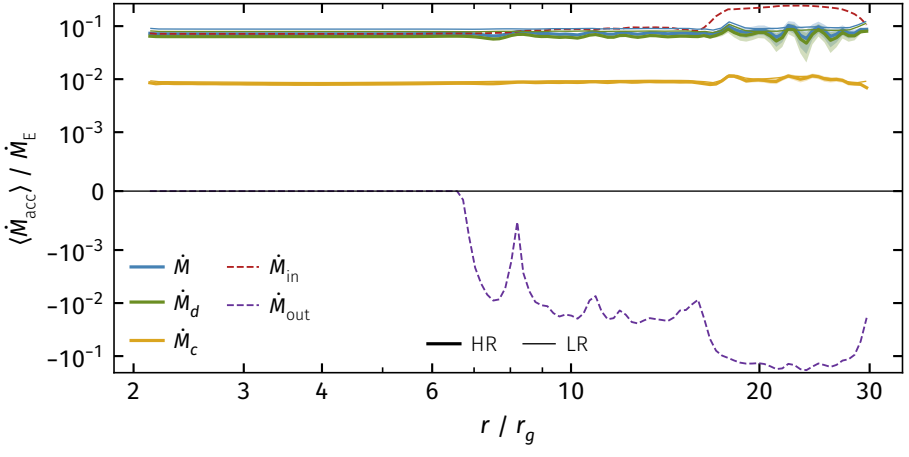
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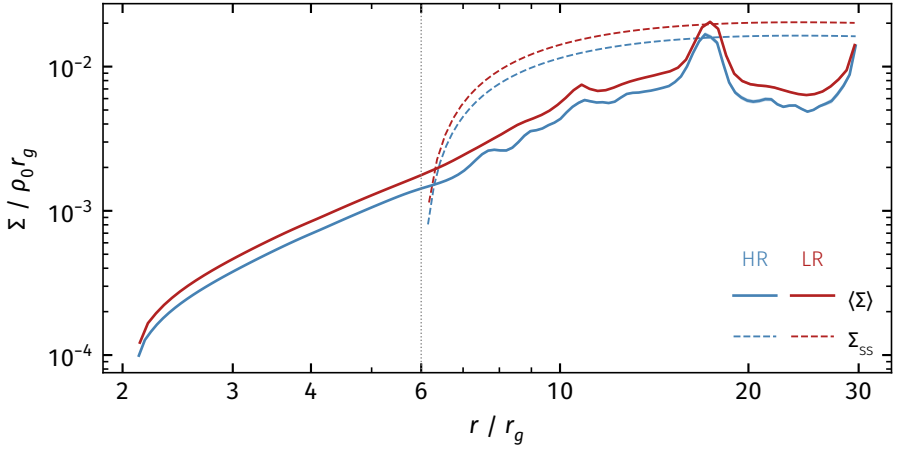
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  - Gas a Hundred to Two Thousand Times Hotter than Radiation at Midplane

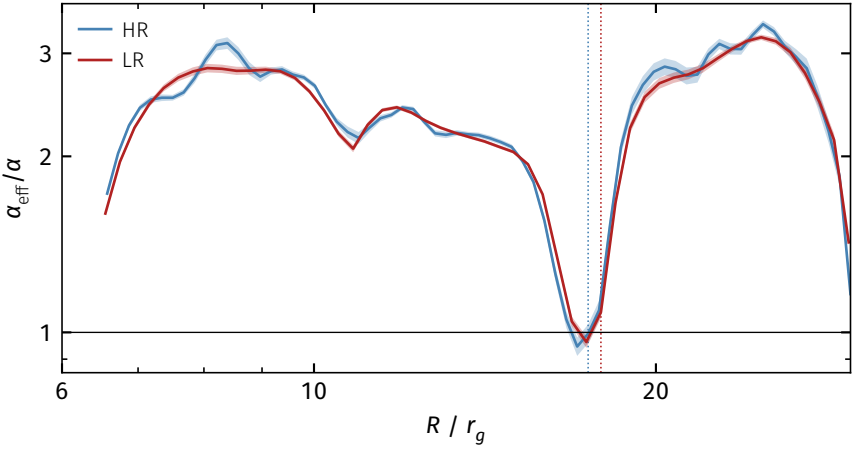
Steady State – Accretion



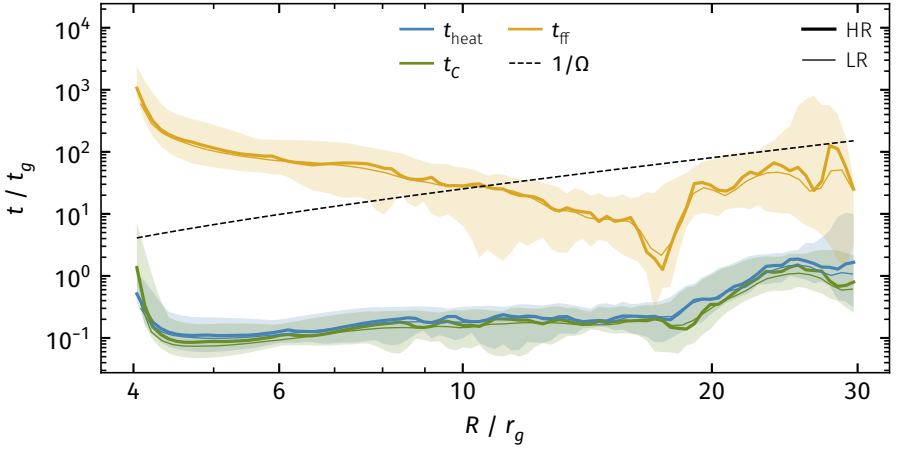
Steady State – Surface Density



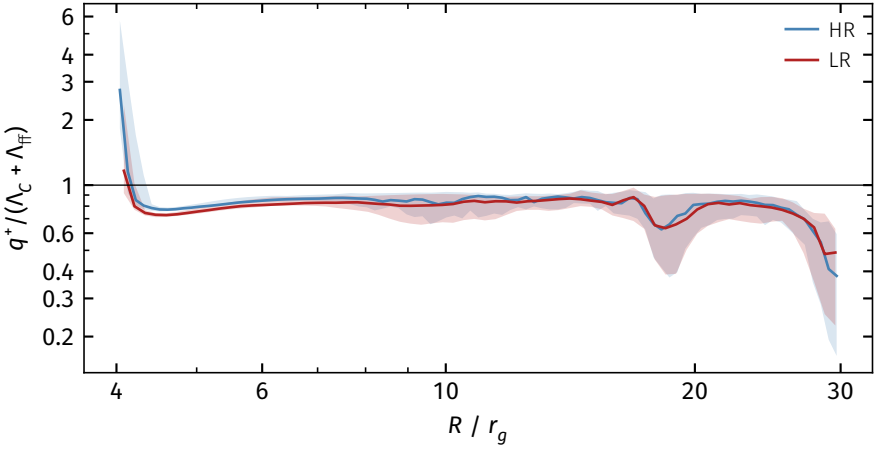
Steady State – Effective Viscosity



Steady State – Heating & Cooling Times



Steady State – Heating & Cooling Ratio



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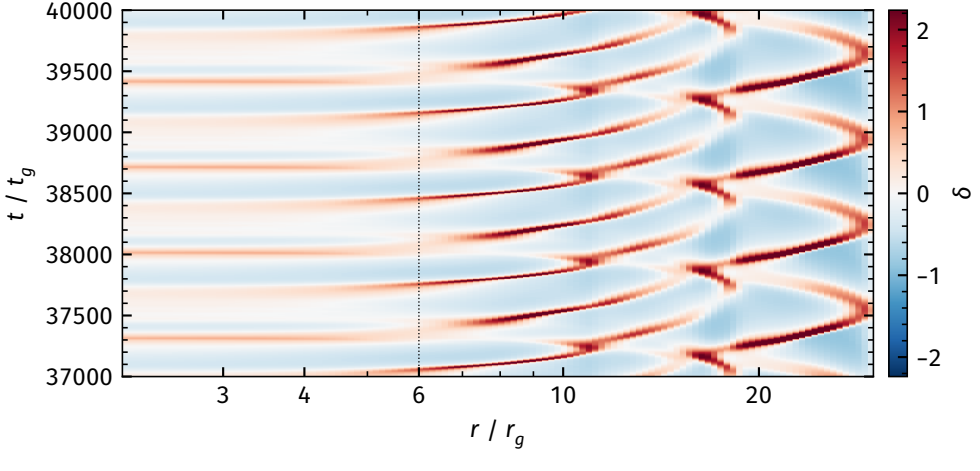
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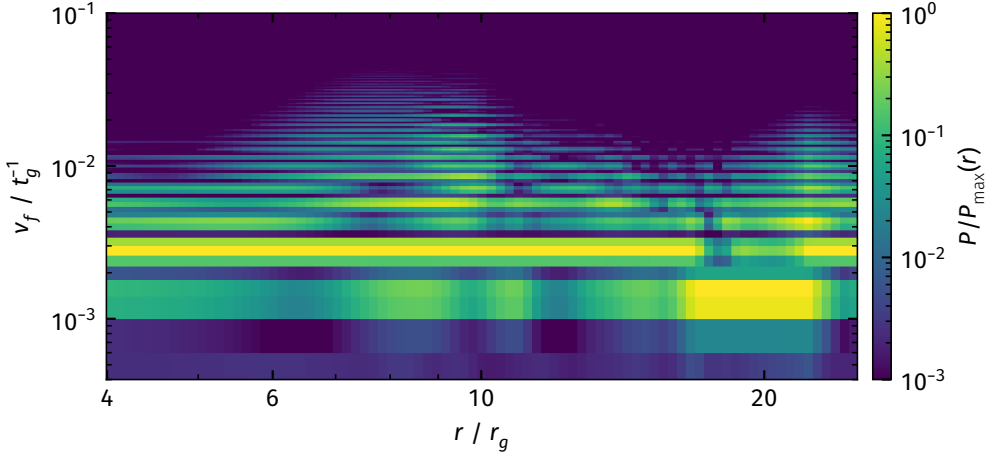
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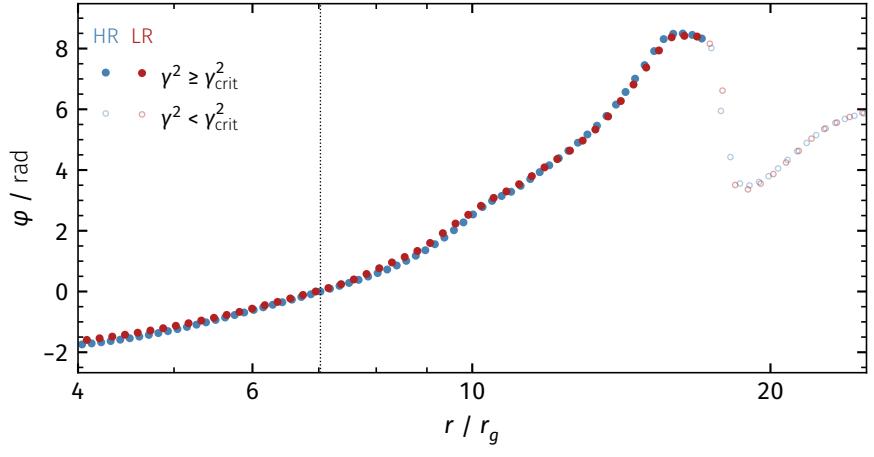
Density Waves – Perturbation



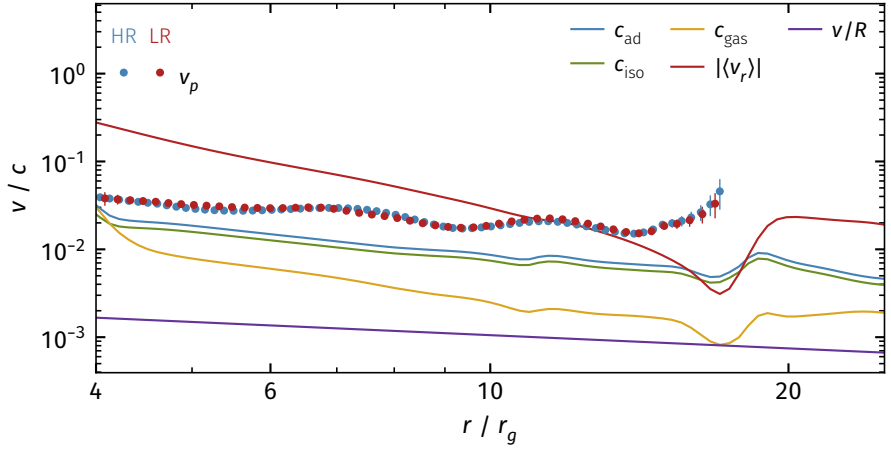
Density Waves – Power Spectrum



Density Waves – Phase



Density Waves – Phase Speed



Resolution Comparison

Quantity	HR	LR
Cells ($r \times \theta$)	120×90	90×45
Classifier Version	Current	First
Steady State Start / t_g	23 725	9225
Accretion Rate / $\dot{M}_E$	0.073	0.091
Luminosity / L_E	0.056	0.070
Efficiency	0.048	0.049
Equilibrium Radius / r_g	17.4	17.9
Aspect Ratio ($R = 10$)	0.057	0.042
Cells per Scale Height	3.3	1.2
Wave Period / t_g	349	349
Phase Speed / c	0.0190	0.0195

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  - Nonlinear Density Waves with Period of about **350** t_g Traveling Outward

Thank You!

- A. Mignone et al., *PLUTO: A Numerical Code for Computational Astrophysics* (ApJS) 2007
- M. Čemeljić, *Atlas of Numerical Solutions for Star-Disk Magnetospheric Interaction* (A&A) 2019
- J. D. Melon Fuksman et al., *A Two-Moment Radiation Hydrodynamics Scheme Applicable to Simulations of Planet Formation in Circumstellar Disks* (ApJ) 2021
- B. Paczyński, P. J. Wiita, *Thick Accretion Disks and Supercritical Luminosities* (A&A) 1980
- Y.-F. Jiang et al., *Super-Eddington Accretion Disks around Supermassive Black Holes* (ApJ) 2019

Resources on GitHub: here

Model Equations

- Initial Disk Pressure and Density from Analytic Solution of Kluźniak & Kita

$$C = \frac{2}{5\epsilon^2}\left(\frac{1}{r} - \frac{1 - 5\epsilon^2/2}{R}\right) \qquad p = \epsilon^2 C^{5/2} \qquad \rho = C^{3/2}$$

- Corona in Hydrostatic Equilibrium with $\rho_c = 0.01$ in Code Units

$$\rho = \rho_c r^{-3/2} \qquad p = 0.4\rho_c r^{-5/2}$$

- Shear Viscosity with Sound Speed of Initial Disk, Restricted by Classifier

$$\eta_v = \frac{2}{3}\alpha\rho\epsilon^2 C R^{3/2} f \qquad v = \frac{2}{3}\alpha\epsilon^2 R^{1/2}$$

- Density Floor at 5×10^{-7} in Code Units with Constant Sound Speed and Momentum

Opacity Model

- Kramers Free Free and Bound Free Absorption with Guillotine Factor

$$\kappa_K = \left(K_{\rm ff} + K_{\rm bf}\right) \rho\, T_g^{-7/2} \qquad \kappa_{\rm es} = 0.2\,(1 + X)$$

- Compton Exchange Held at Cap above $T_{\rm max} = 1 \times 10^9$ K

$$\kappa_C = \kappa_{\rm es} \frac{4 k_B T_r^4 \left(T_{\rm max} - T_r\right)}{m_e c^2 \left(T_g^4 - T_r^4\right)}$$

- Gated by Classifier and Decoupled near Speed of Light, Thomson Scattering on All Gas

$$f_d = \max\left(\frac{f - 0.05}{0.95}, 0\right) \qquad g_v = \left[1 + \exp\left(\frac{\beta - 0.9}{0.03}\right)\right]^{-1}$$

$$\kappa_{\rm abs} = g_v f_d \left(\kappa_K + \kappa_C\right) \qquad \kappa_{\rm sc} = g_v \left(\kappa_{\rm es} - f_d \kappa_C\right)$$

Classifier Details

- Corona Reference as $1 - f$ and Disk Reference as f Weighted Mean Density per Bin
  - Disk Estimate Counts Only above One Thousandth of Bin Volume
  - Bins without Valid Estimate Copy Nearest Valid Bin
- Guards Added after Inversion in First Version
  - Span below 0.5 dex Copies Both References from Nearest Sufficient Bin
  - Disk Reference Raised to Minimum Span if No Such Bin Exists
  - Fixed Weight 0.05 per Step Replaced by Time Constant τ
- Optional Geometric Mode Independent of Classification
  - Disk Reference from Maximum Density, Corona from Geometric Mean above $z/R = 0.5$

Analysis Methods

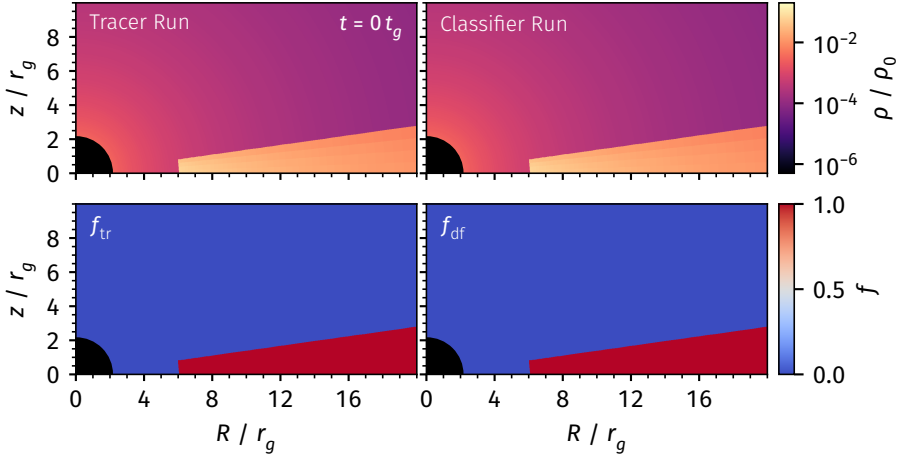
- Standard Errors from Block Averages over **1000** Gravitational Times
- Relaxation from Running Means over $W = 2500$ Compared with Final Fifth

$$D(t_c) = \frac{|\bar{X}_W(t_c) - X_{\mathrm{ref}}|}{\sigma}$$

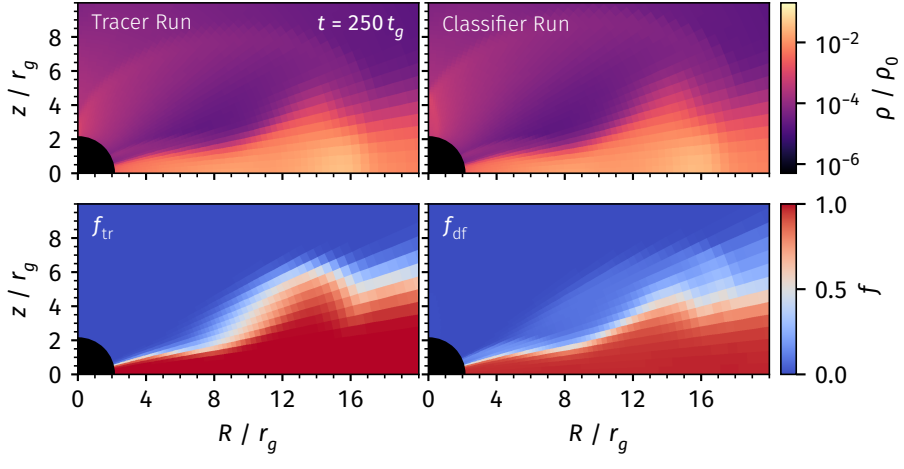
- Inflow Equilibrium where Accretion Rate within **20 %** of Inner Value
- Welch Spectra of Surface Density Perturbation $\delta = \Sigma/\langle\Sigma\rangle - 1$
  - Phase at Peak Frequency Relative to $r = 7$ for Significant Coherence Only
  - Local Phase Speed from Linear Fits over **15 %** in Radius

$$v_p = \omega \,/\, \frac{\mathrm{d}\varphi}{\mathrm{d}r}$$

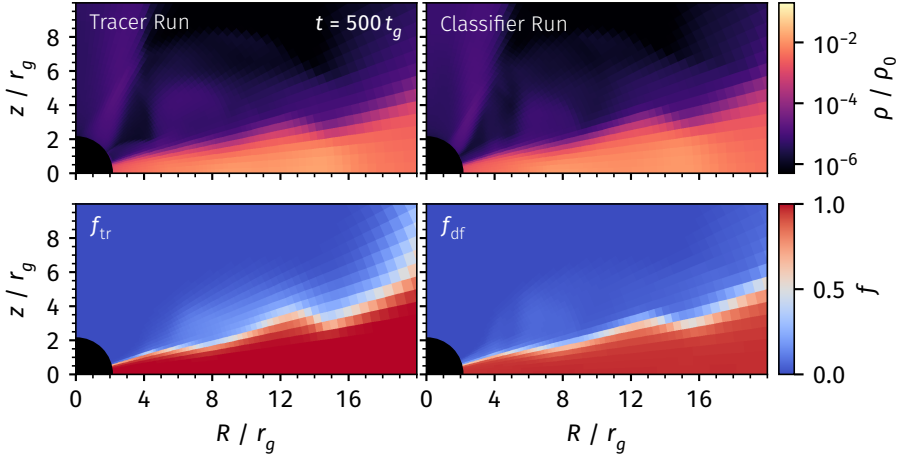
Validation – Magnetohydrodynamics



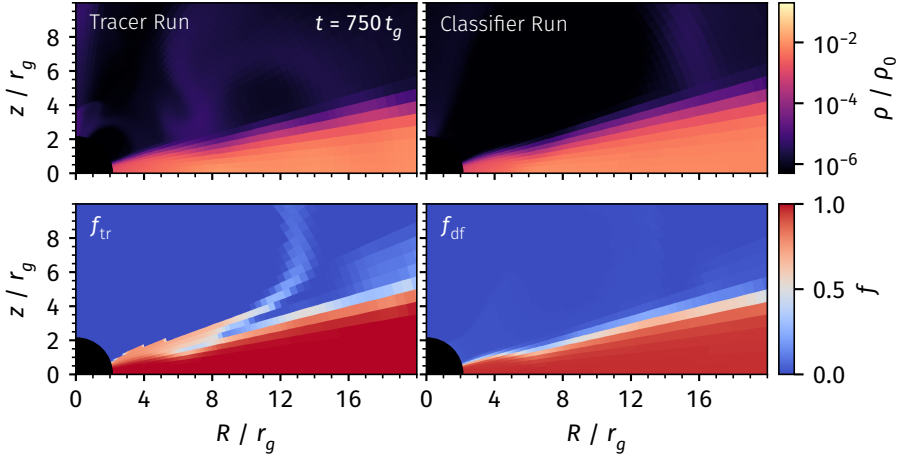
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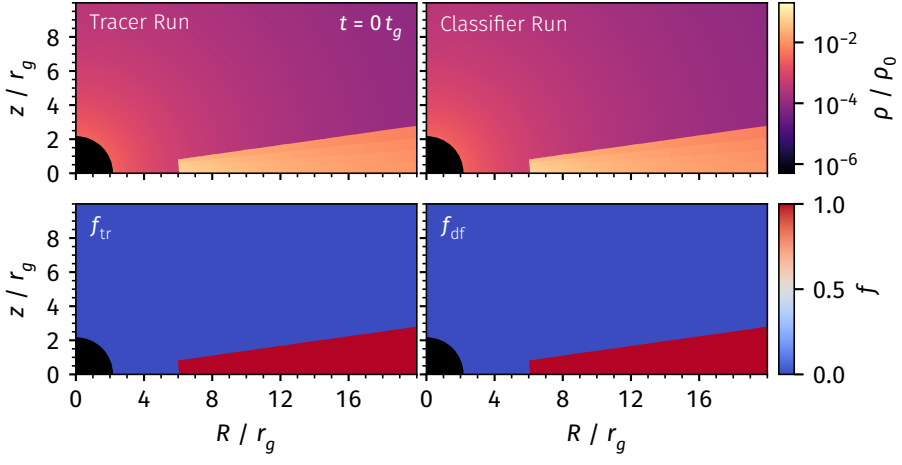
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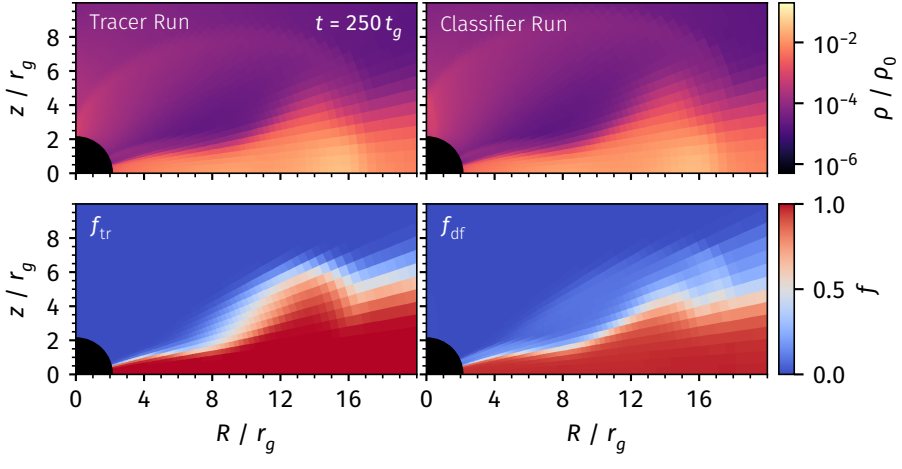
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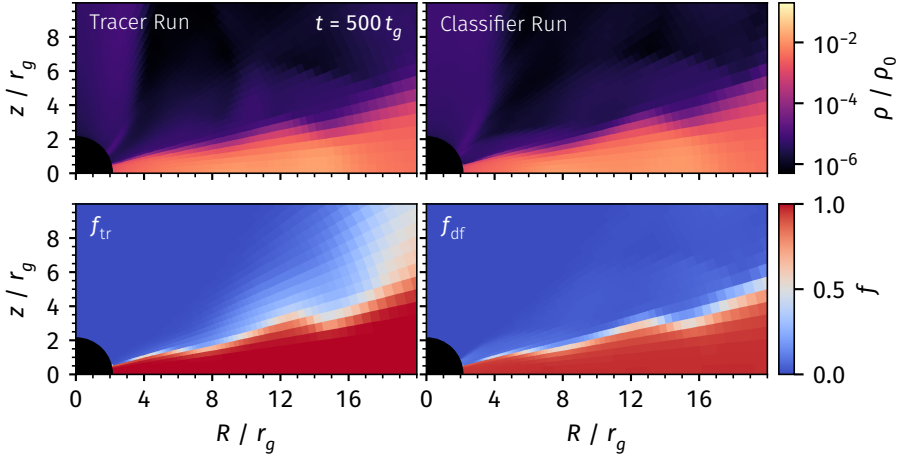
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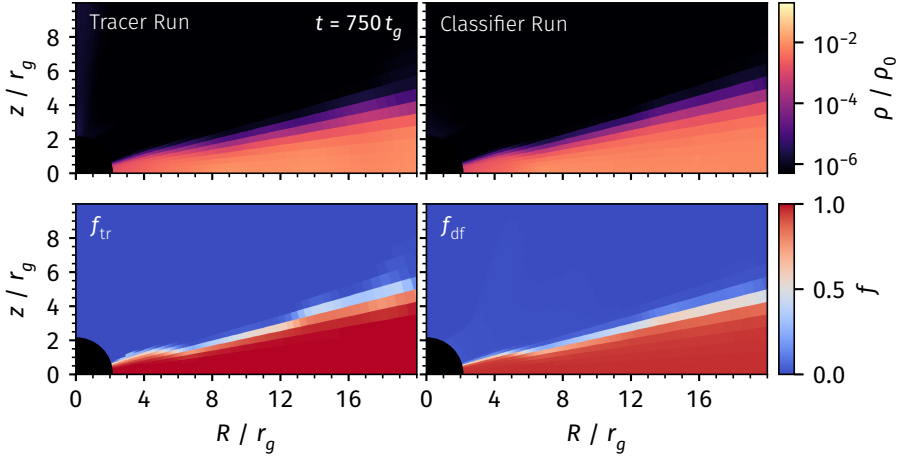
Validation – Resistive Magnetohydrodynamics



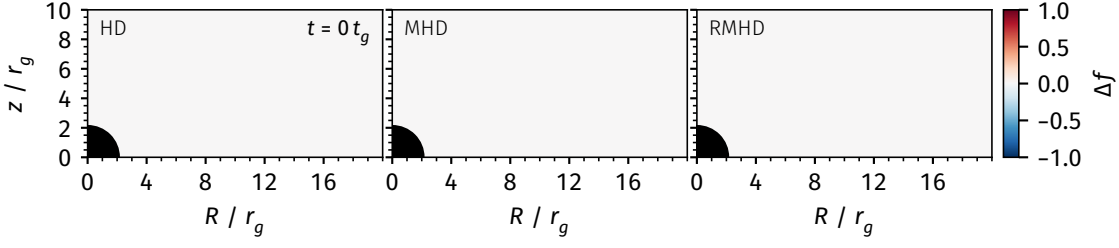
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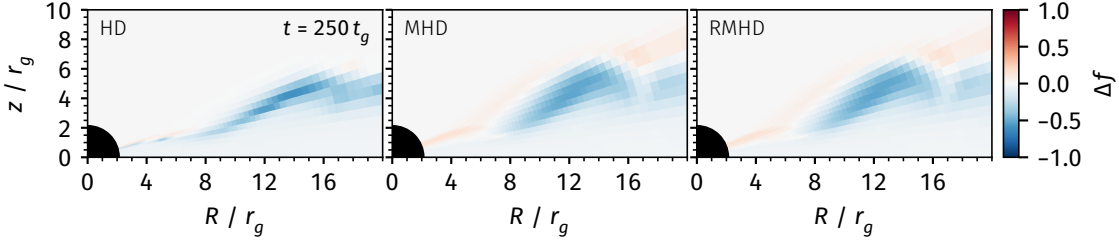
Validation – Resistive Magnetohydrodynamics



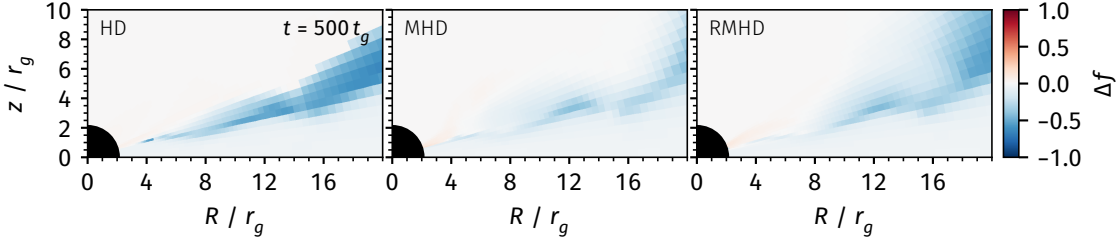
Validation – Classifier Minus Tracer



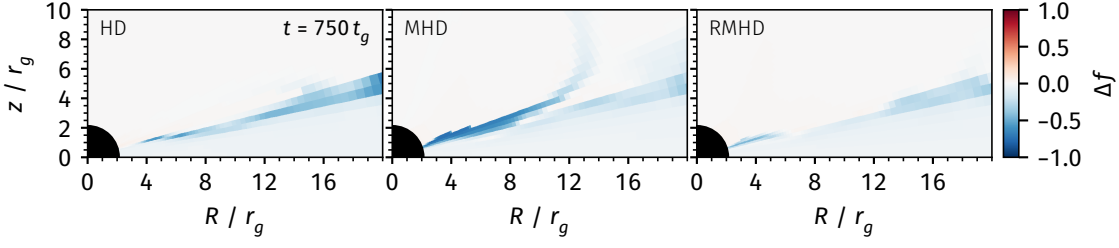
Validation – Classifier Minus Tracer



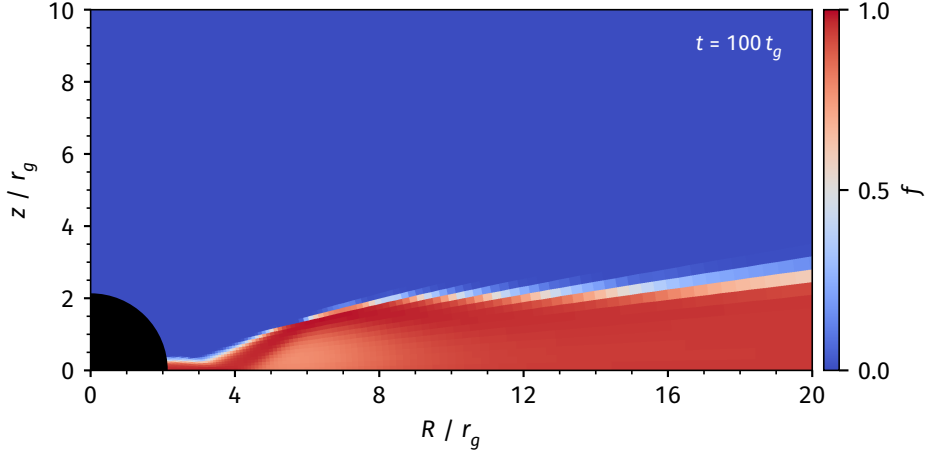
Validation – Classifier Minus Tracer



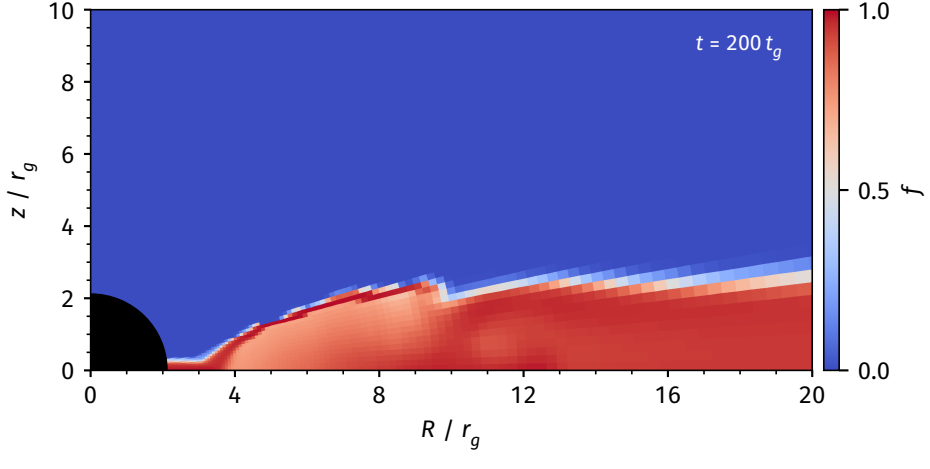
Validation – Classifier Minus Tracer



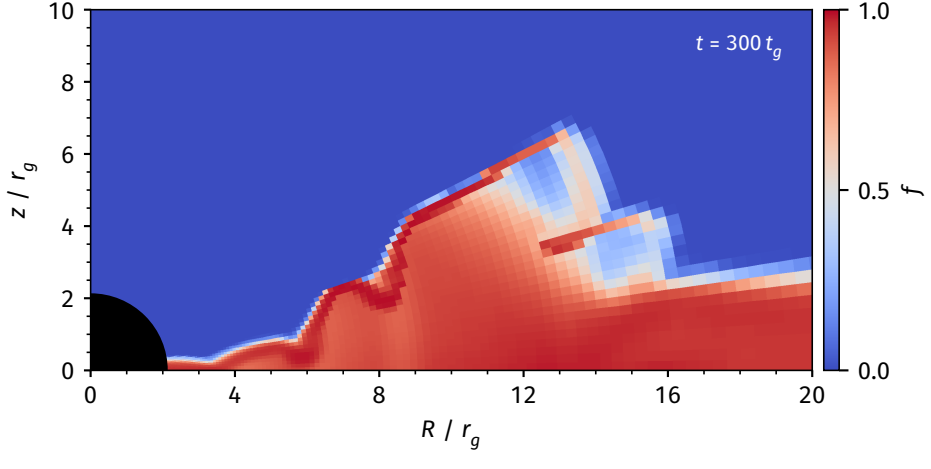
Early Evolution – DiskFraction



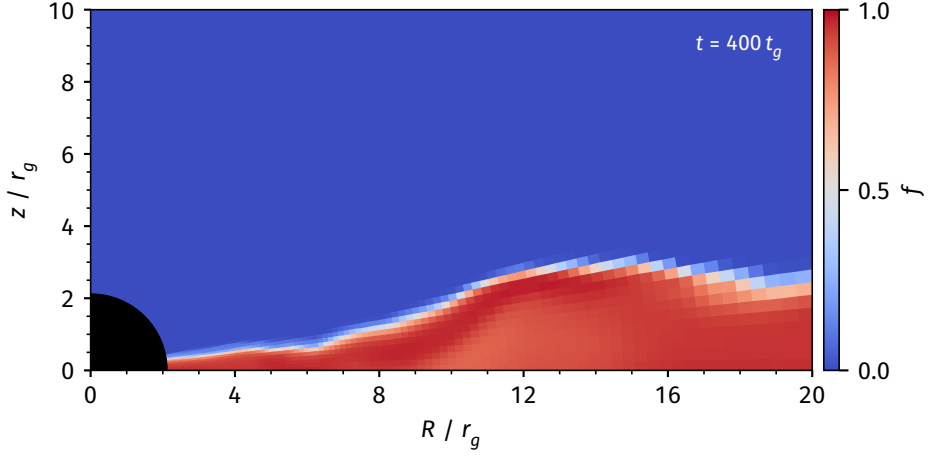
Early Evolution – DiskFraction



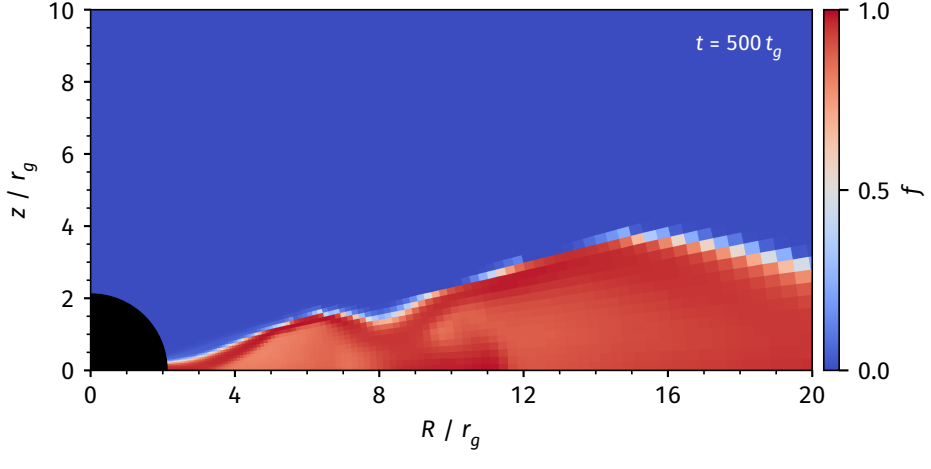
Early Evolution – DiskFraction



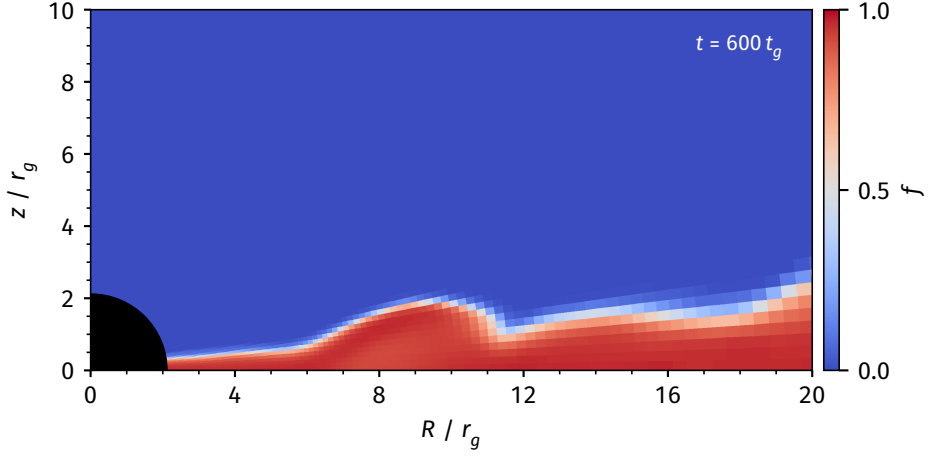
Early Evolution – DiskFraction



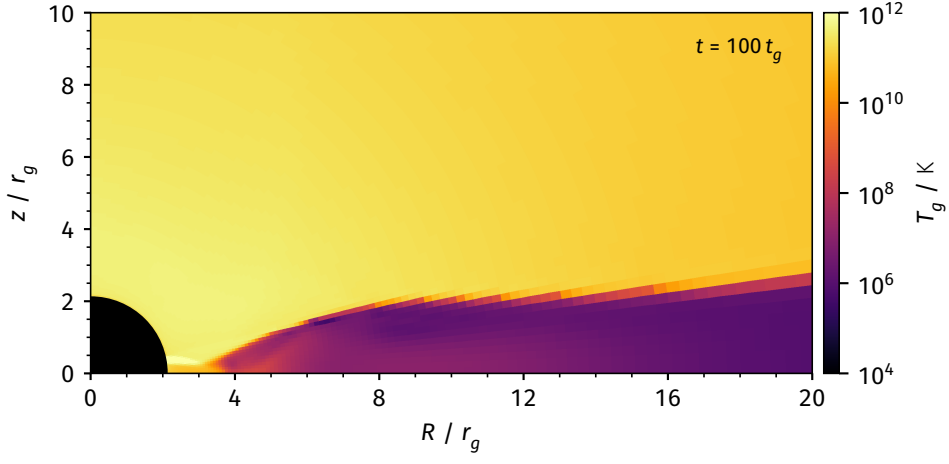
Early Evolution – DiskFraction



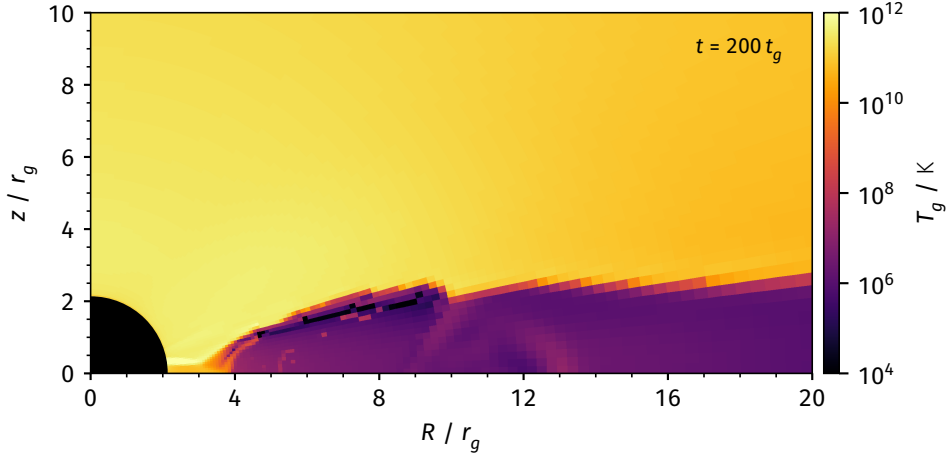
Early Evolution – DiskFraction



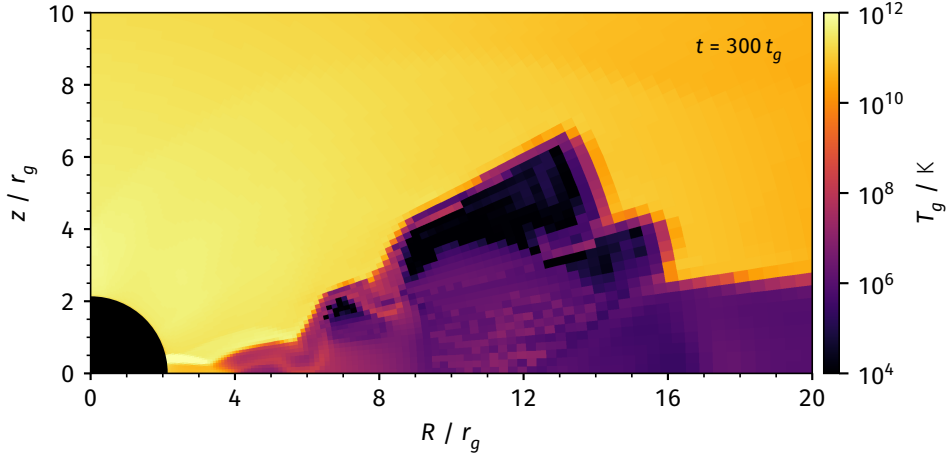
Early Evolution – Gas Temperature



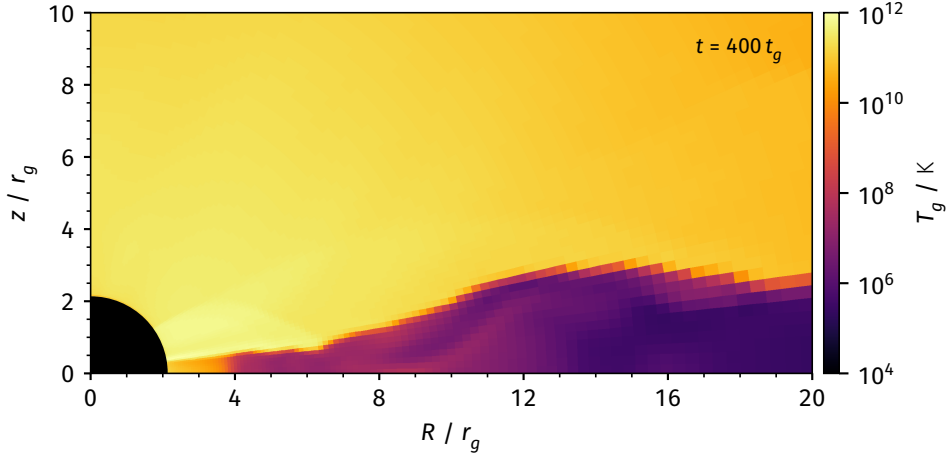
Early Evolution – Gas Temperature



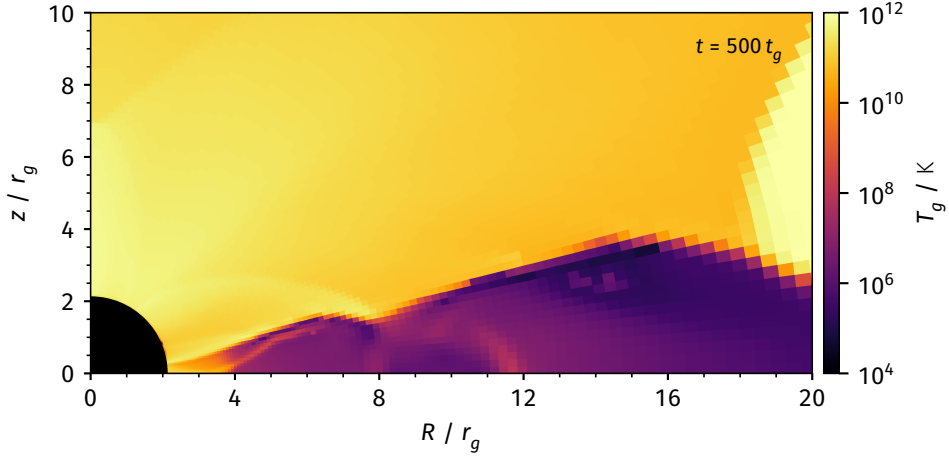
Early Evolution – Gas Temperature



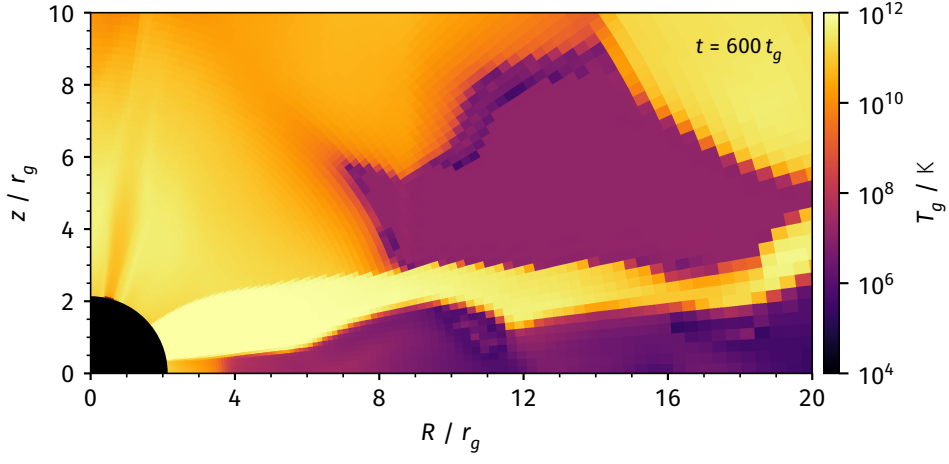
Early Evolution – Gas Temperature



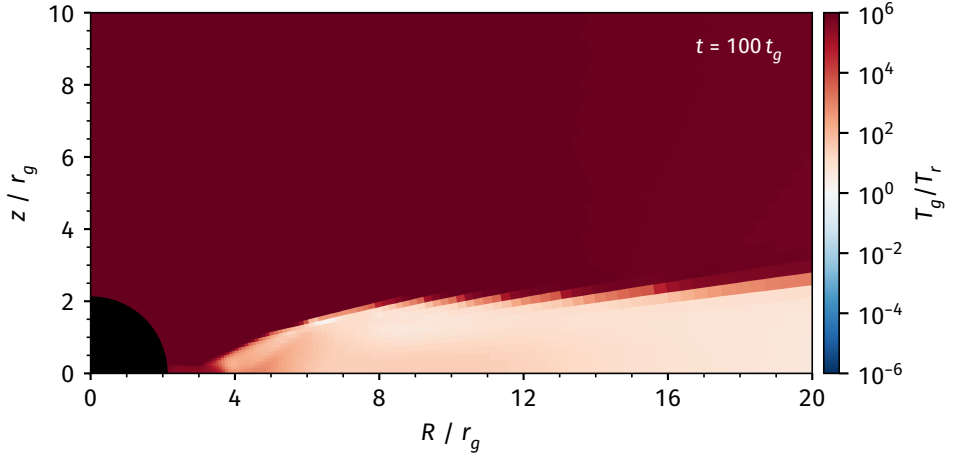
Early Evolution – Gas Temperature



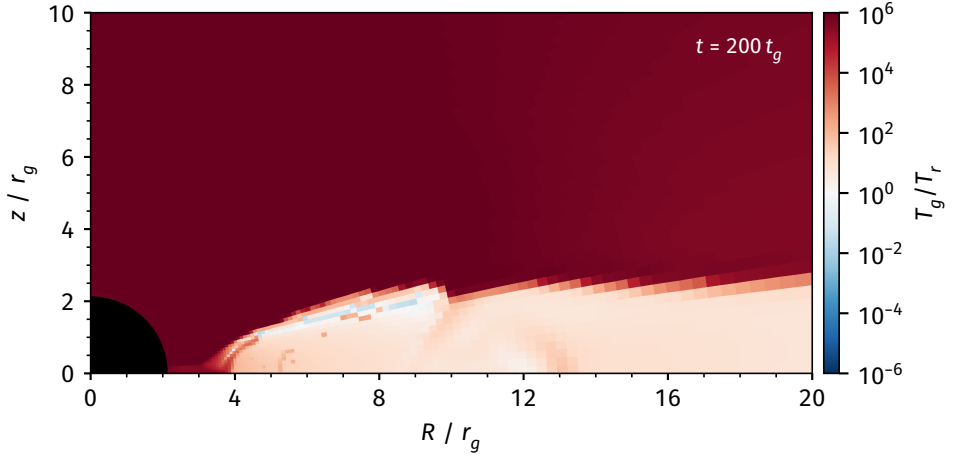
Early Evolution – Gas Temperature



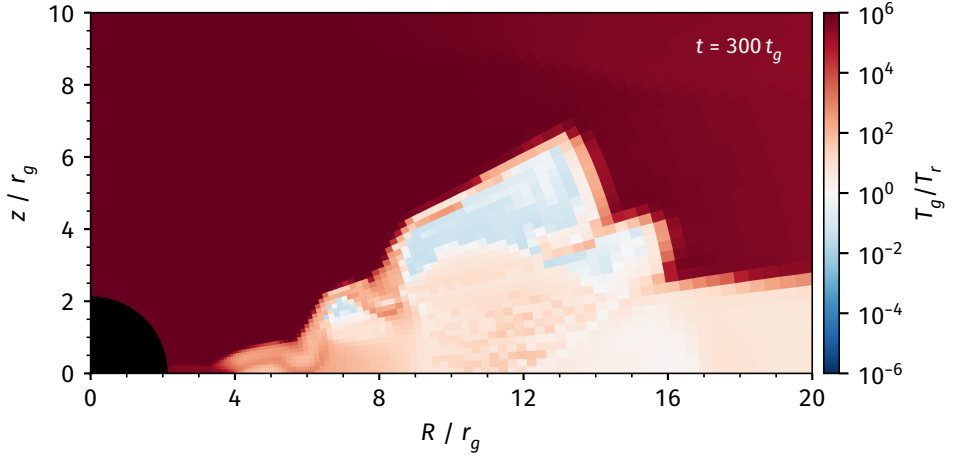
Early Evolution – Temperature Ratio



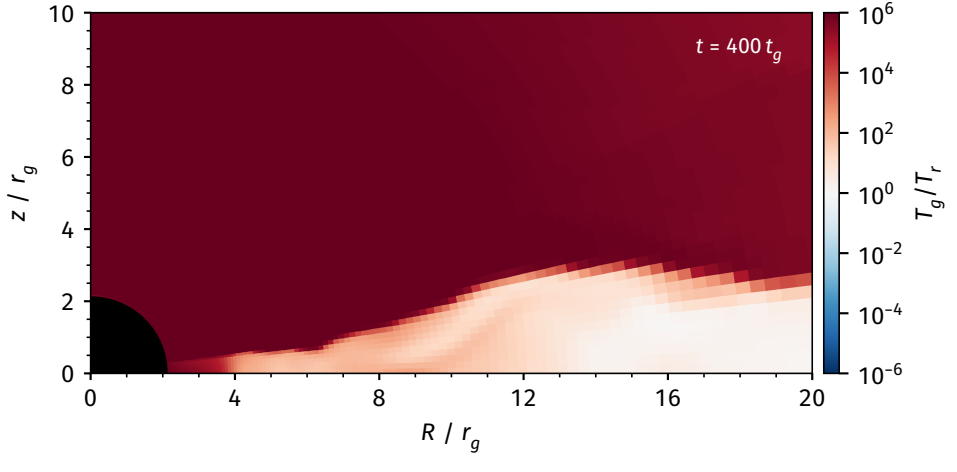
Early Evolution – Temperature Ratio



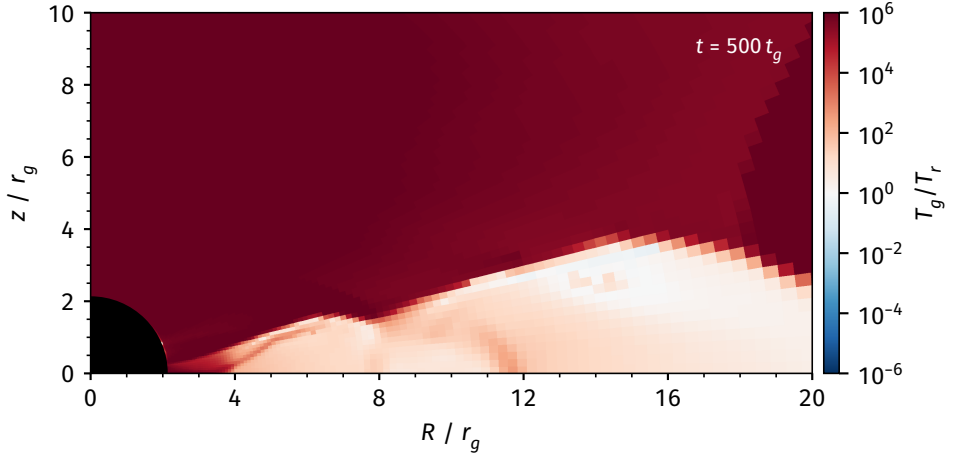
Early Evolution – Temperature Ratio



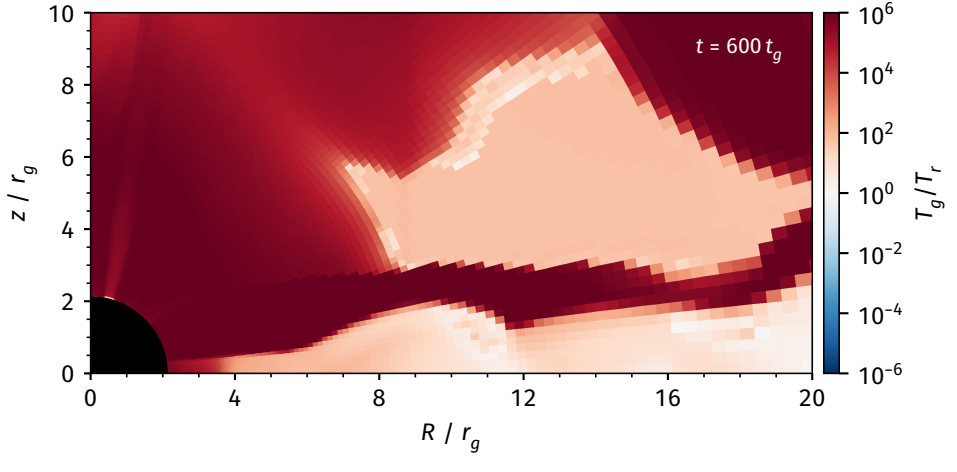
Early Evolution – Temperature Ratio



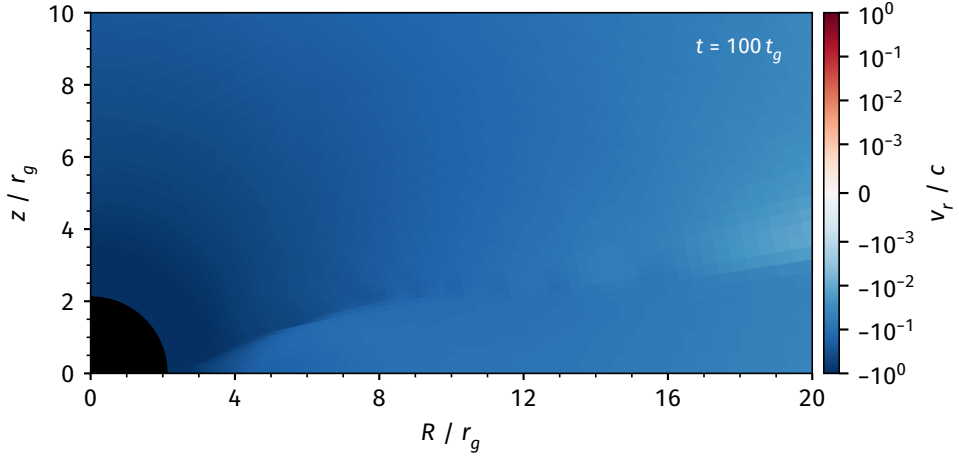
Early Evolution – Temperature Ratio



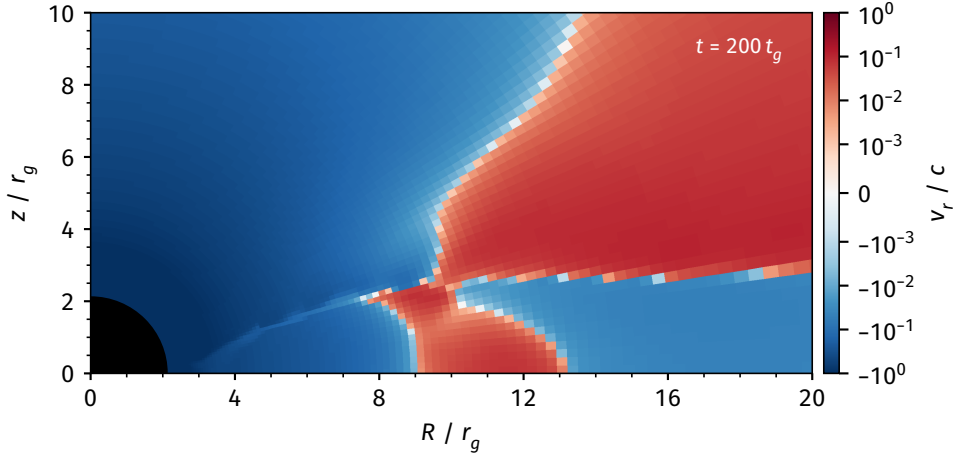
Early Evolution – Temperature Ratio



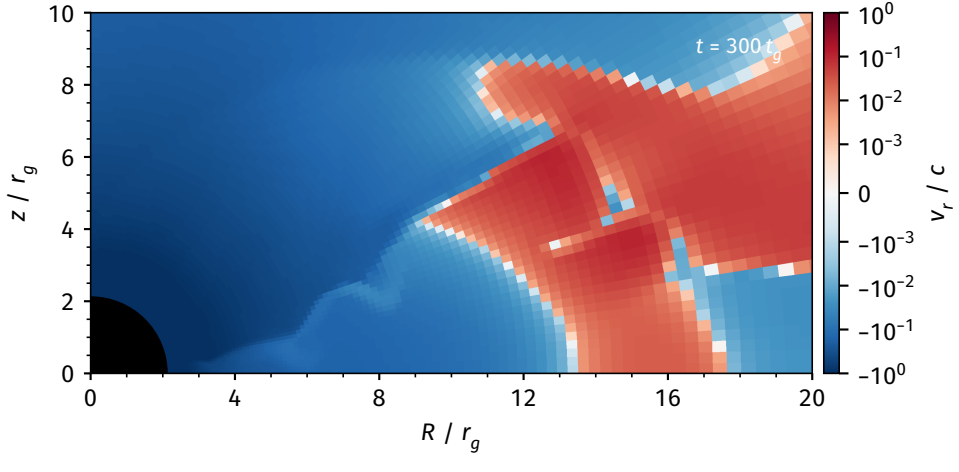
Early Evolution – Radial Velocity



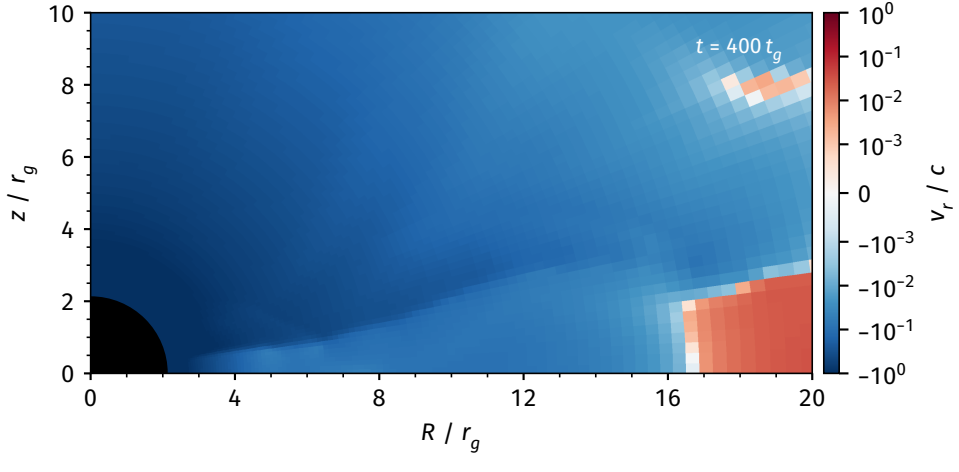
Early Evolution – Radial Velocity



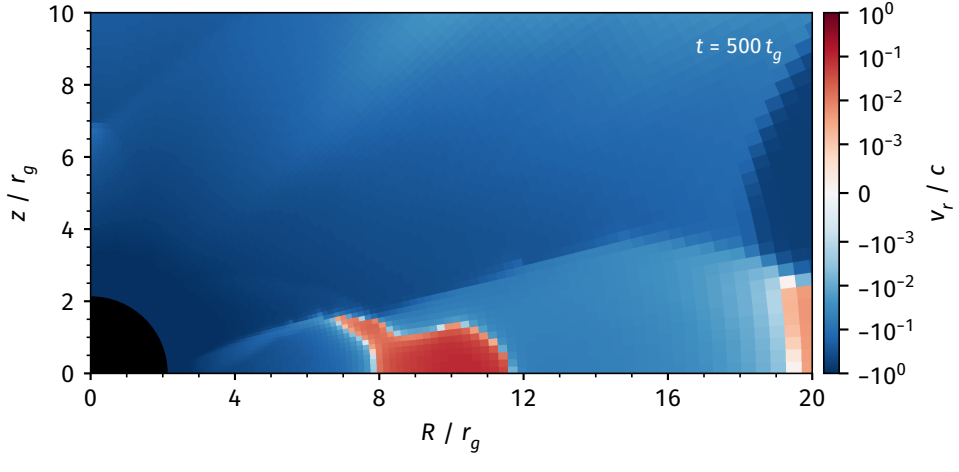
Early Evolution – Radial Velocity



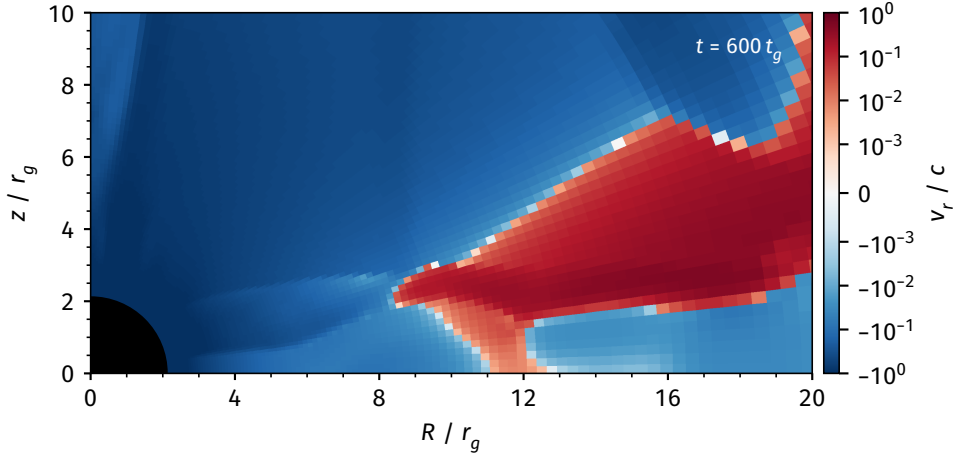
Early Evolution – Radial Velocity



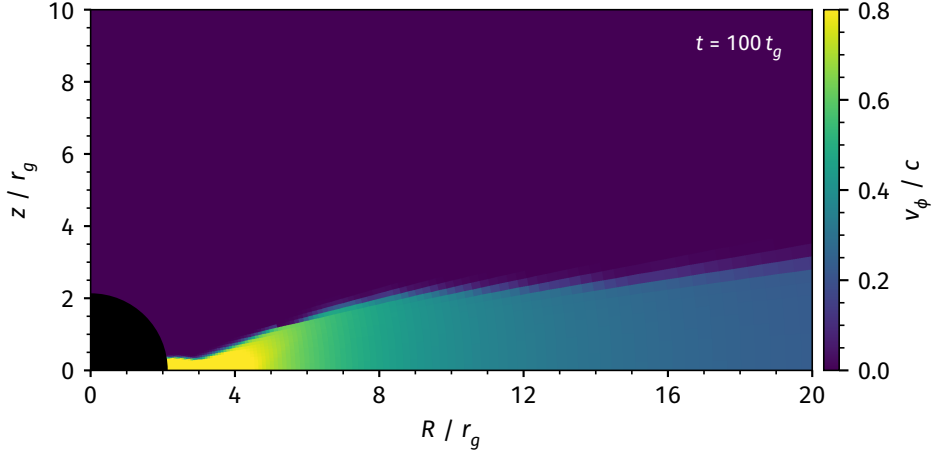
Early Evolution – Radial Velocity



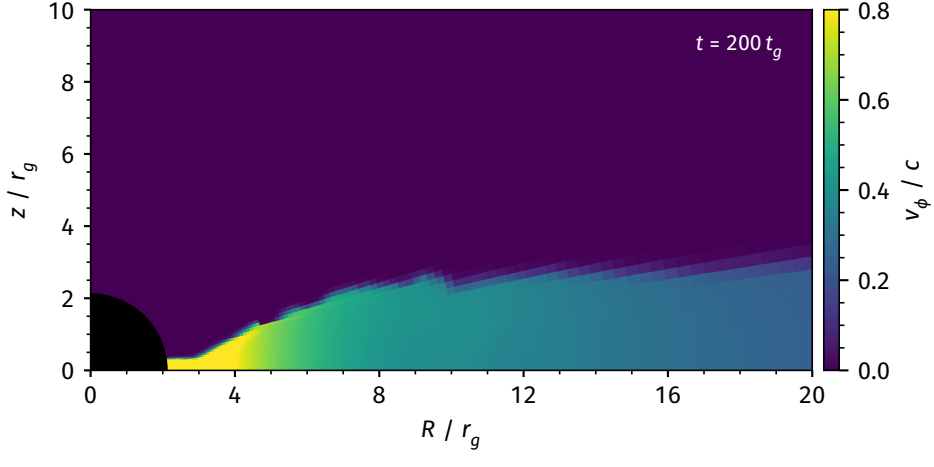
Early Evolution – Radial Velocity



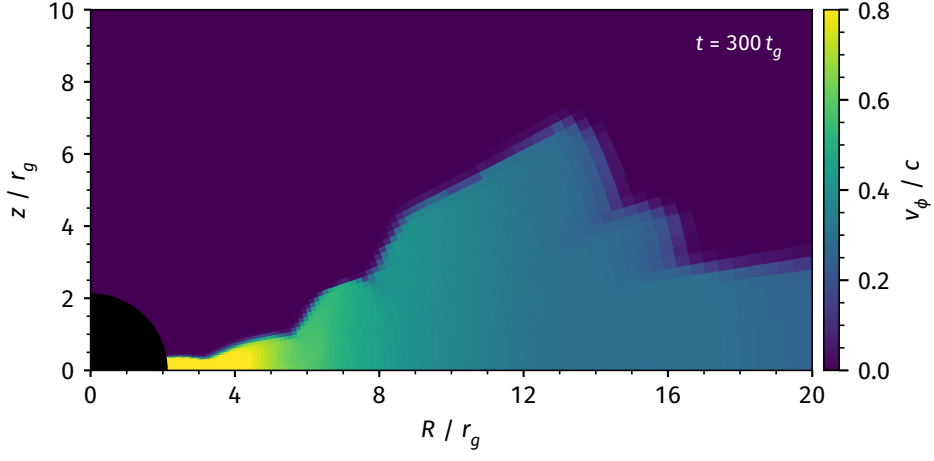
Early Evolution – Rotation Velocity



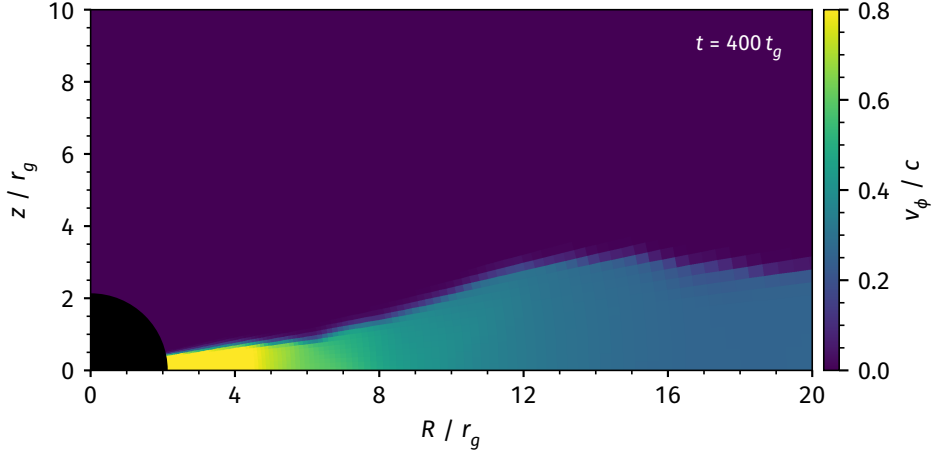
Early Evolution – Rotation Velocity



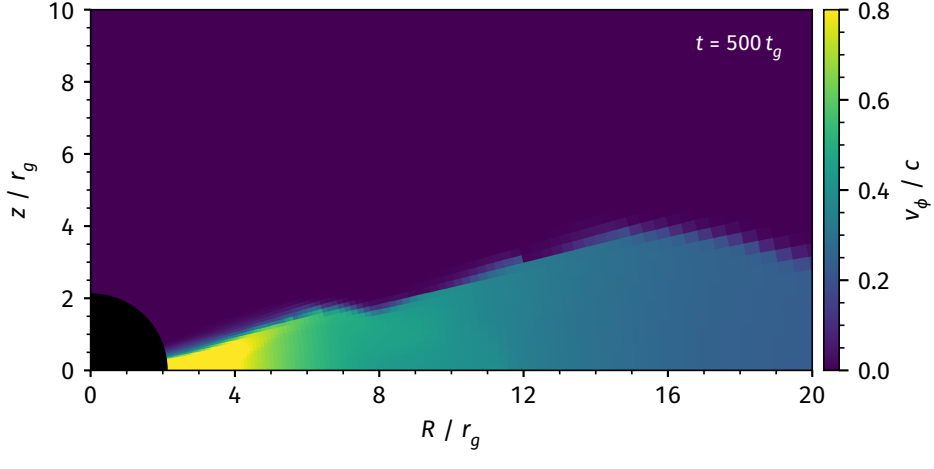
Early Evolution – Rotation Velocity



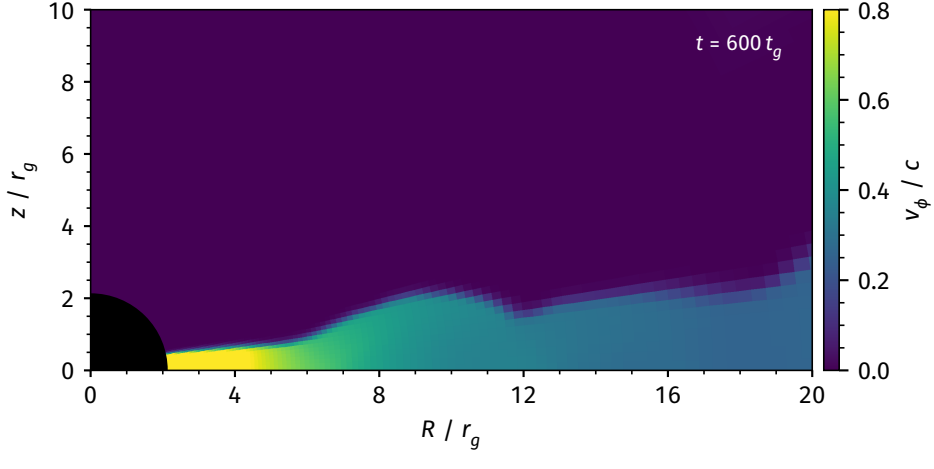
Early Evolution – Rotation Velocity



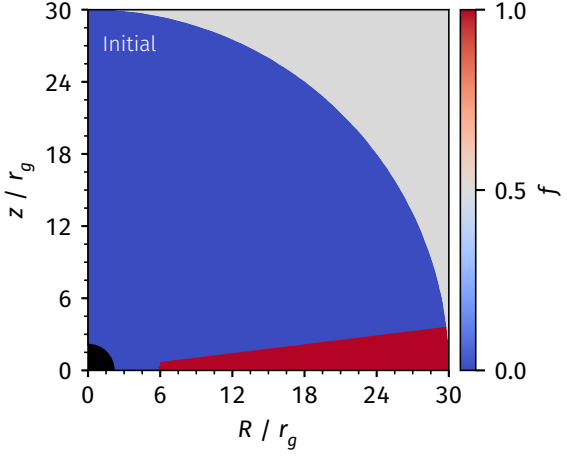
Early Evolution – Rotation Velocity



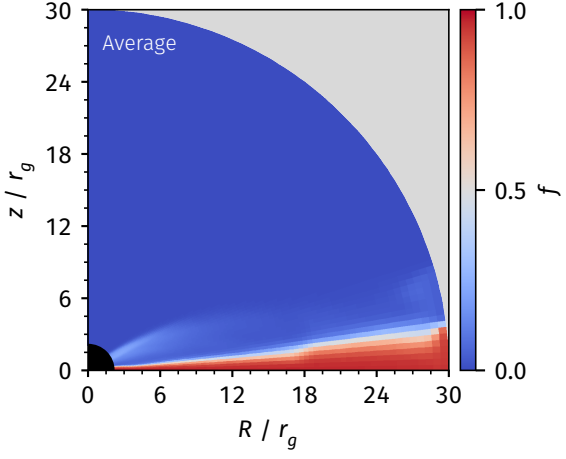
Early Evolution – Rotation Velocity



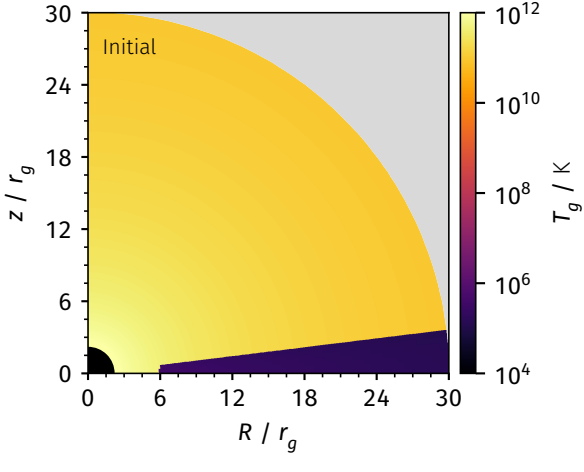
Steady State – DiskFraction



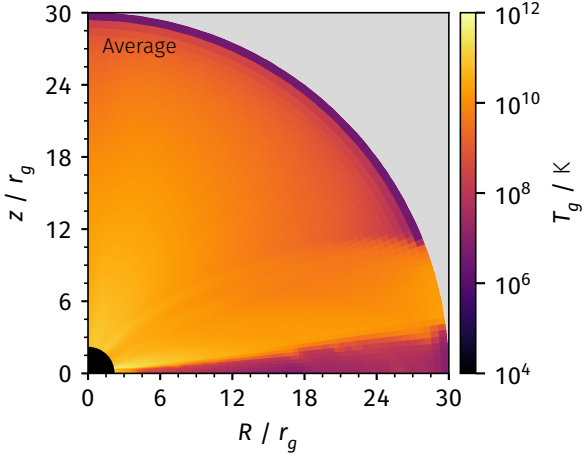
Steady State – DiskFraction



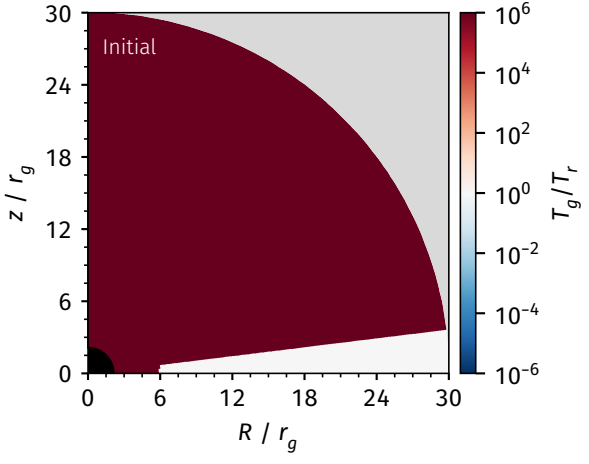
Steady State – Gas Temperature



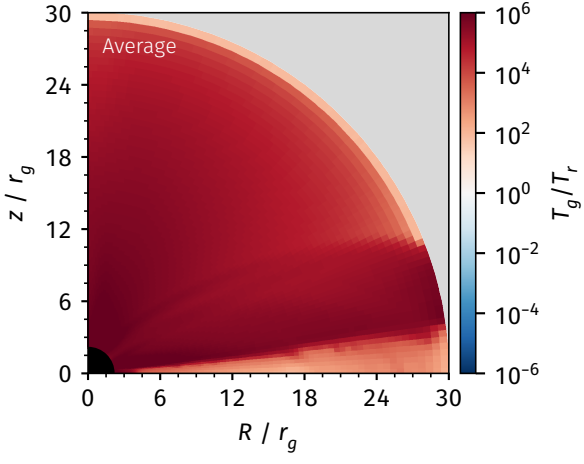
Steady State – Gas Temperature



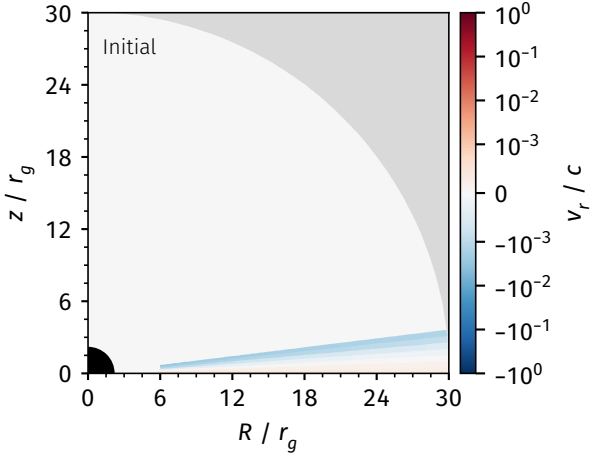
Steady State – Temperature Ratio



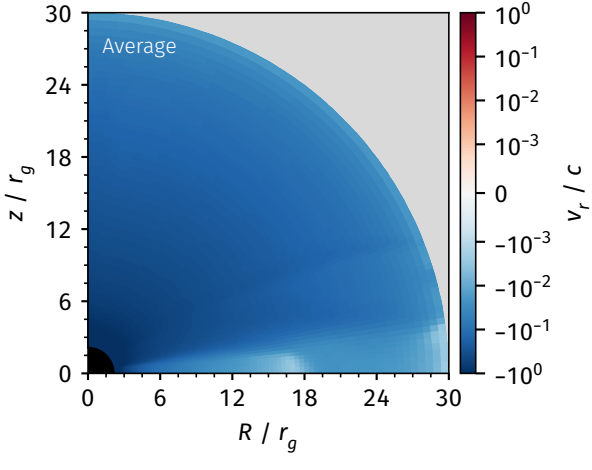
Steady State – Temperature Ratio



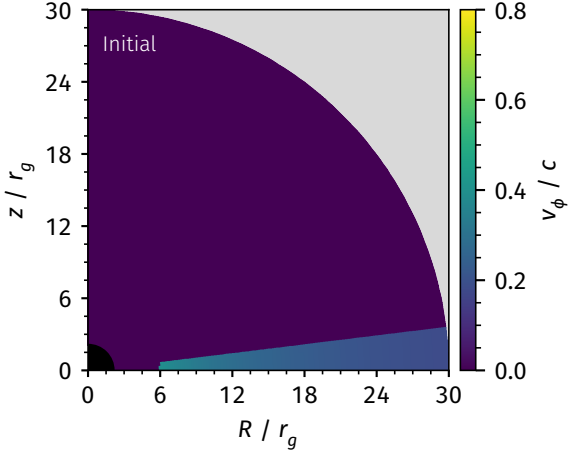
Steady State – Radial Velocity



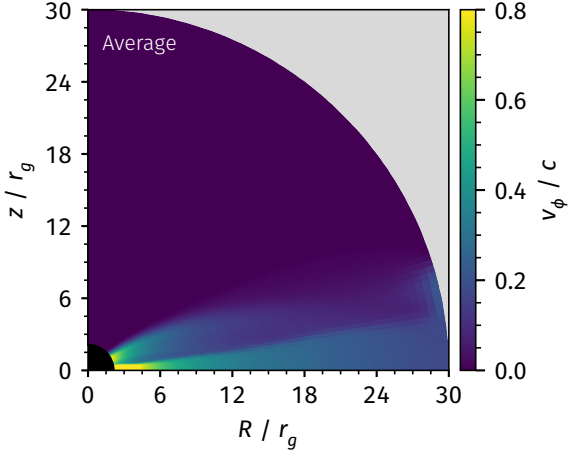
Steady State – Radial Velocity



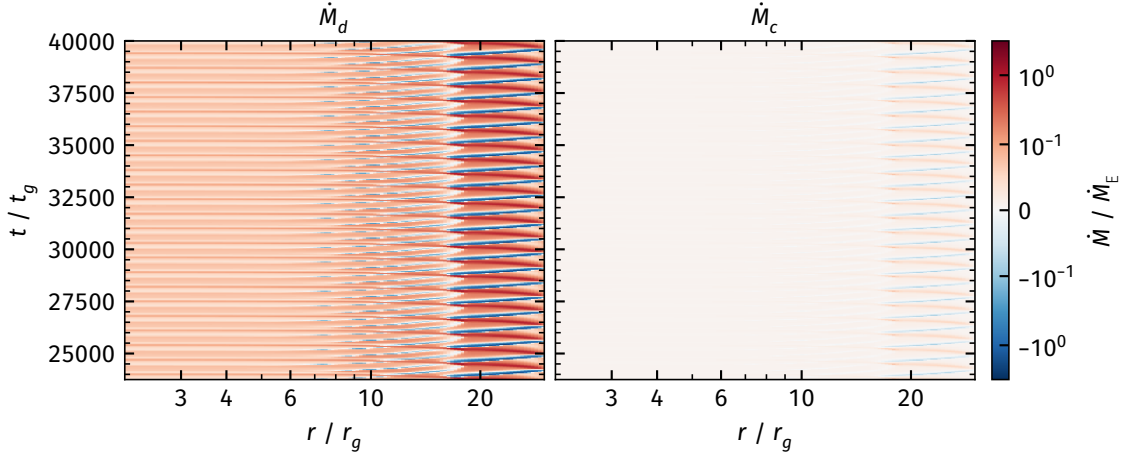
Steady State – Rotation Velocity



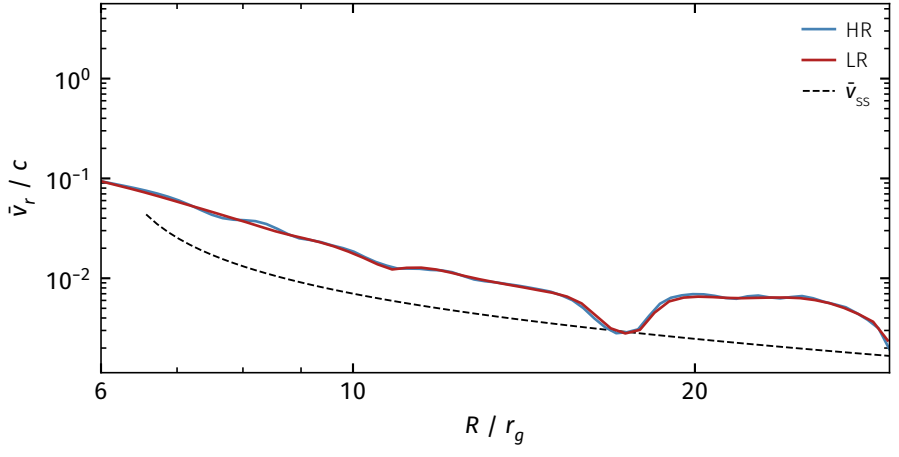
Steady State – Rotation Velocity



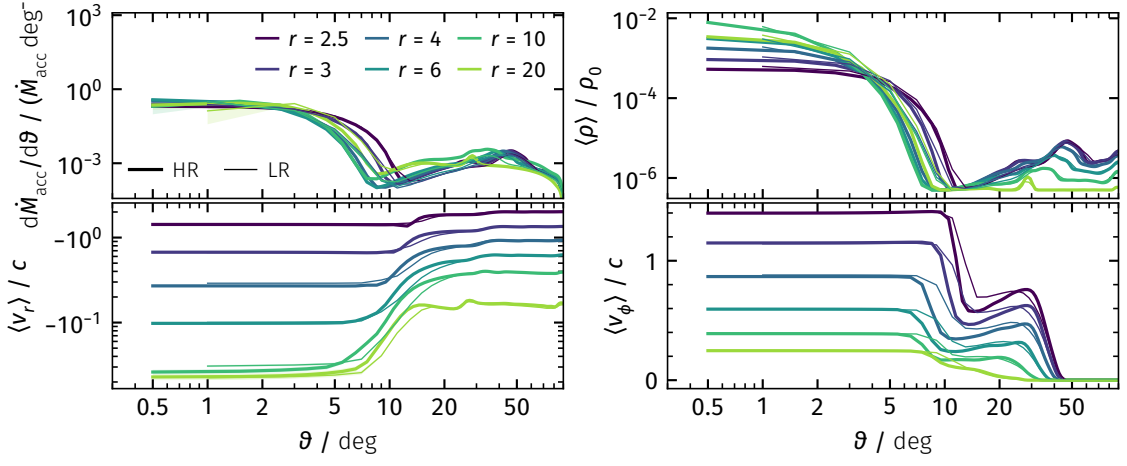
Steady State – Accretion Rate Map



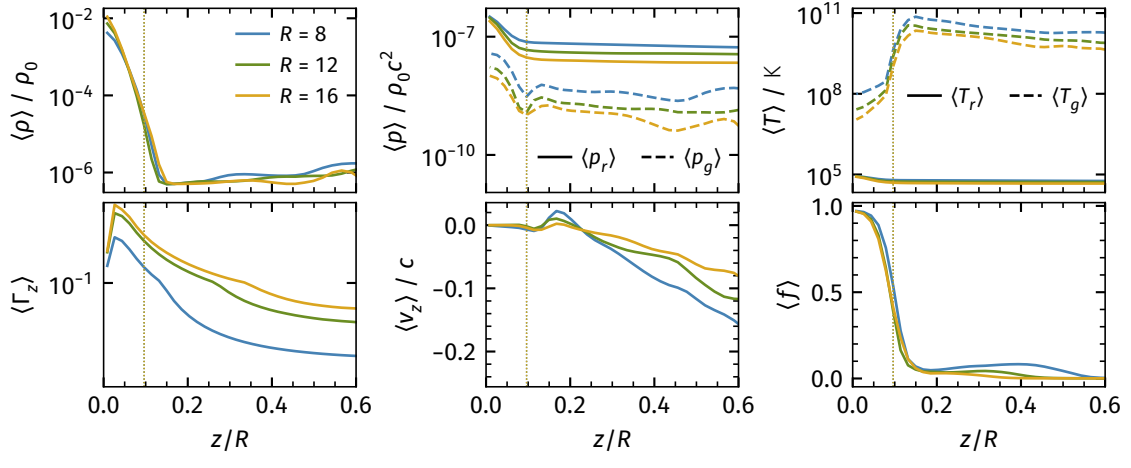
Steady State – Inflow Speed



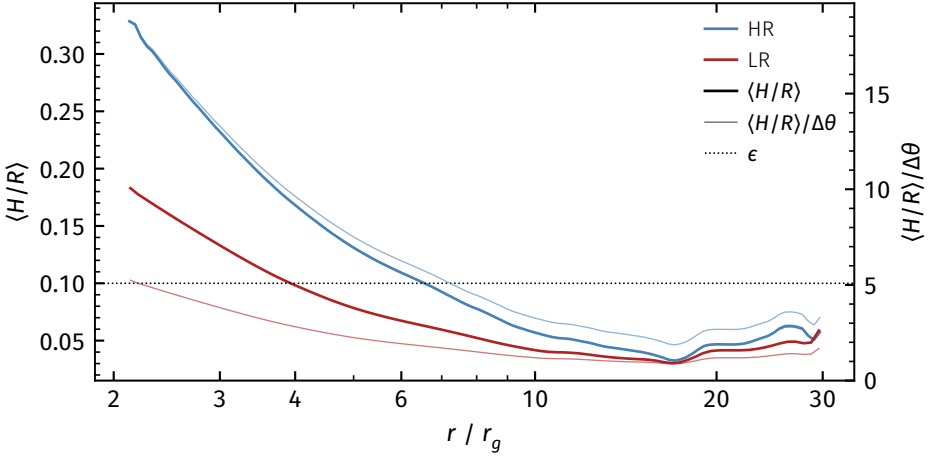
Steady State – Angular Profiles



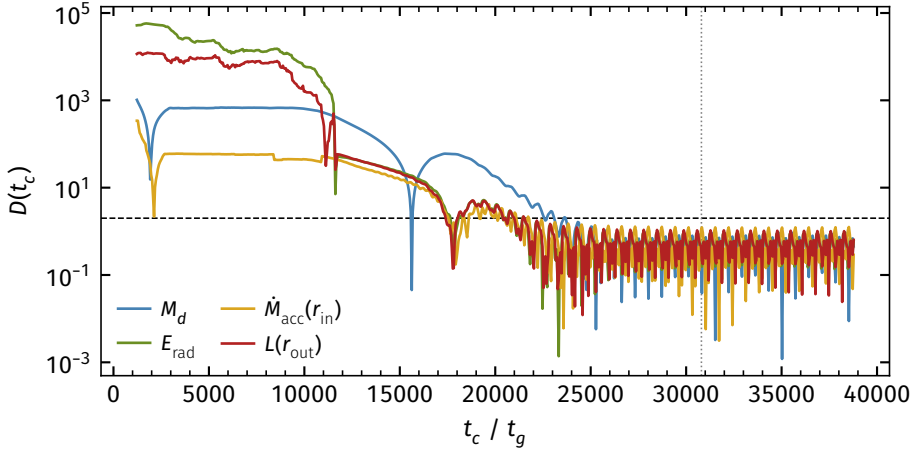
Steady State – Vertical Profiles



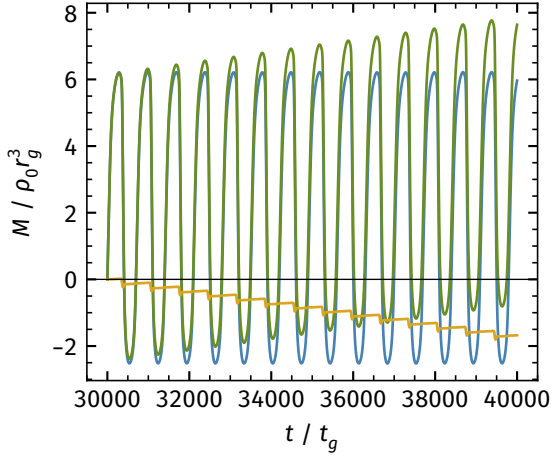
Steady State – Aspect Ratio

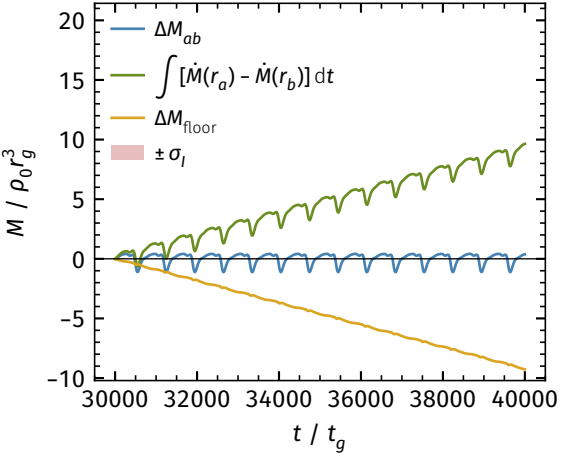


Diagnostics – Relaxation Statistic

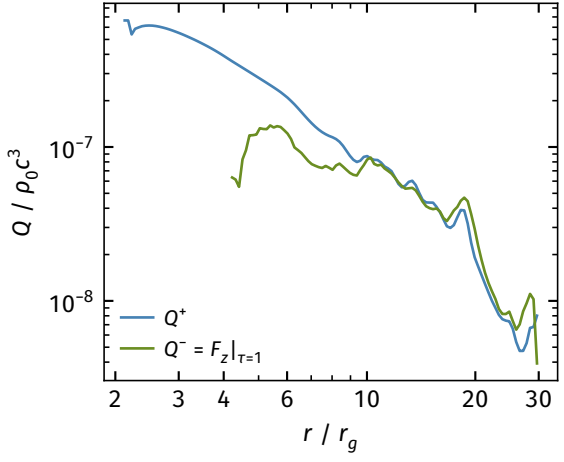


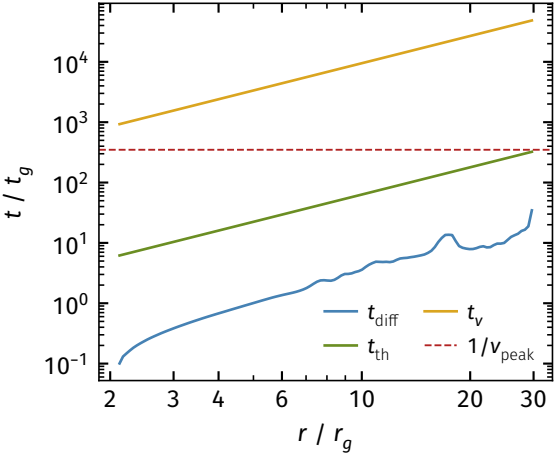
Diagnostics – Mass Budget



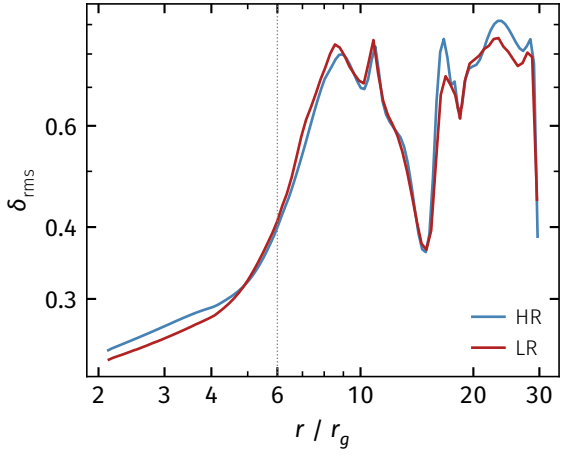


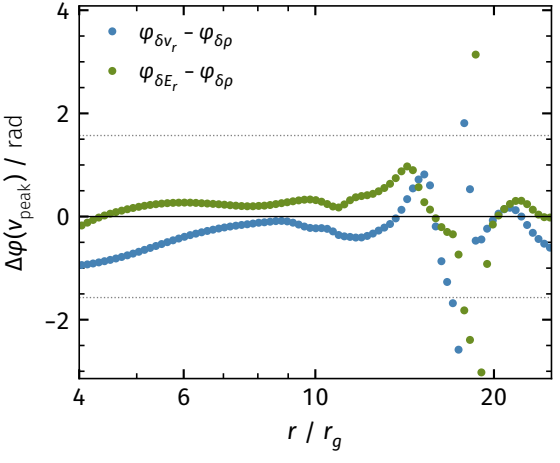
Diagnostics – Energy Balance





Diagnostics – Wave Amplitude & Cross Phase





Diagnostics – Timescales

